

Supplemental Material for “Spectral Rendering Without a Spectral Renderer: Learned Spectral Codes for RGB Pipelines”

JIAQI YU, University of York, United Kingdom

DAR’YA GUARNERA, University of York, United Kingdom

GIUSEPPE CLAUDIO GUARNERA, University of York, United Kingdom and Lumirithmic Ltd, United Kingdom

This supplemental document expands what the main paper references but does not develop in full: the algebraic basis of the codec design, the architecture and fidelity of the learned codec, the training and ablation protocols behind the proposed configuration, the capacity analysis that motivates $k=6$, the material-class encoding details for glass, conductors, and participating media, and the design and evaluation of the RGB upsampling networks. We also include a narrowband-illumination evaluation, a wall-clock timing comparison against a modern spectral renderer (Mitsuba 3, with hero wavelength sampling), and the complete per-case results of the cross-renderer evaluation.

CCS Concepts: • **Computing methodologies** → **Rendering**; *Reflectance modeling*; *Dimensionality reduction and manifold learning*; *Neural networks*.

Additional Key Words and Phrases: spectral rendering, spectral compression, RGB rendering pipelines, multi-pass rendering, spectral upsampling, Hadamard product, metamerism

What is in this supplemental

Each section anchors to a specific main-paper claim and reports the additional detail or evidence that does not fit in the body.

Sec. 1 establishes that no exact $k < n$ compression of the per-channel-product spectral algebra exists, so the codec must be approximate; this grounds the loss design of Sec. 3.

Sec. 2 details the codec architecture and quantifies what it preserves, under direct viewing (Sec. 2.3) and under repeated rendered products (Sec. 2.4).

Sec. 3 reports the loss formulae, hyperparameter search, training-data corpora and splits (Sec. 3.3), and the per-component ablation.

Sec. 4 establishes the capacity argument: $k=3$ trades off direct-viewing accuracy against product-preservation (Sec. 4.1); under matched training, learned bases dominate fixed alternatives (Sec. 4.2); alignment with the subspace of the colour-matching functions (CMFs) is necessary but not sufficient (Sec. 4.3); and $k=9$ does not justify its extra rendering pass (Sec. 4.4).

Sec. 5 characterises codec behaviour across a 52-cell narrowband-illumination sweep, locating the basis-gap regions and quantifying the advantage over an RGB baseline.

Sec. 6 covers material-class encoding details for coloured glass, conductors, and homogeneous / heterogeneous subsurface scattering (SSS), i.e. how each Mitsuba plugin’s spectral inputs flow through the

codec; it also holds the extended material-class evaluation (Sec. 6.4) and the per-regime colour-error characterisation (Sec. 6.5).

Sec. 7 describes the RGB → latent upsampling networks for legacy assets without spectral data, and evaluates them: held-out accuracy, spectral fidelity of the uplifted illuminants, and directional error.

Sec. 8 compares wall-clock and quality against Mitsuba 3’s spectral path tracer (hero wavelength sampling).

Sec. 9 gives the ground-truth construction protocol, the per-renderer configurations, and all thirteen per-case scores behind Sec. 6.4 of the main paper.

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Authors’ Contact Information: Jiaqi Yu, jiaqiyu0307@outlook.com, University of York, York, United Kingdom; Dar’ya Guarnera, darya.guarnera@york.ac.uk, University of York, York, United Kingdom; Giuseppe Claudio Guarnera, claudio.guarnera@york.ac.uk, University of York, York, United Kingdom and Lumirithmic Ltd, London, United Kingdom.



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1 Algebraic preliminaries and impossibility proof

No exact algebra-preserving compression exists from n -dimensional spectra to a $k < n$ latent: there is no injective ring homomorphism $E : (\mathbb{R}^n, +, \odot) \rightarrow (\mathbb{R}^k, +, \odot)$ for $k < n$. This forces the codec to satisfy $\mathcal{L}_{\text{code}}=0$ only in distribution rather than identically (Sec. 1.3), which in turn dictates the four-loss training formulation (Sec. 3.1).

The renderer's per-channel arithmetic is the multiplicative structure $(\mathbb{R}^k, +, \odot)$ where $\odot$ denotes the Hadamard (component-wise) product. The spectral arithmetic that a fully spectrally aware pipeline would use is $(\mathbb{R}^n, +, \odot)$ with $n=47$ in our setup (Mitsuba 0.6's grid; see Sec. 3.3). Our codec $E : \mathbb{R}^n \rightarrow \mathbb{R}^k$ routes spectral computations through the lower-dimensional algebra, so the question is whether E can be a ring homomorphism, i.e. whether $E(s_R \odot s_L) = E(s_R) \odot E(s_L)$ holds for every s_R, s_L . If so, the rendering pipeline composes without error. If not, the identity holds only approximately, and we must characterise that approximation.

1.1 Idempotents in $(\mathbb{R}^n, +, \odot)$

An element e is idempotent if $e \odot e = e$. In $(\mathbb{R}^n, +, \odot)$ this means $e_i^2 = e_i$ for every coordinate, hence $e_i \in \{0, 1\}$. The idempotents of $(\mathbb{R}^n, +, \odot)$ are therefore exactly the 2^n vertices of the unit hypercube $\{0, 1\}^n$.

Two idempotents e, e' are *orthogonal* if $e \odot e' = 0$. In $(\mathbb{R}^n, +, \odot)$ this means their coordinate supports are disjoint. The maximum number of pairwise-orthogonal nonzero idempotents is therefore exactly n , attained by the standard basis $\{e_1, \dots, e_n\}$ where e_i has 1 in position i and 0 elsewhere. This count equals the *rank* of $(\mathbb{R}^n, +, \odot)$ as a finite-dimensional commutative semisimple $\mathbb{R}$ -algebra; the algebra is isomorphic to $\mathbb{R} \times \dots \times \mathbb{R}$ (n copies) and decomposes (Wedderburn) into n simple factors [Atiyah and Macdonald 1969].

1.2 Theorem and proof

Theorem (impossibility). *There is no injective ring homomorphism $\varphi : (\mathbb{R}^n, +, \odot) \rightarrow (\mathbb{R}^k, +, \odot)$ when $k < n$.*

SKETCH. A ring homomorphism preserves products: $\varphi(x \odot y) = \varphi(x) \odot \varphi(y)$. In particular, φ sends idempotents to idempotents and pairwise-orthogonal idempotents to pairwise-orthogonal idempotents.

Let $\{e_1, \dots, e_n\} \subset \mathbb{R}^n$ be the n pairwise-orthogonal nonzero idempotents from Sec. 1.1. By injectivity of φ , $\{\varphi(e_1), \dots, \varphi(e_n)\} \subset \mathbb{R}^k$ are n *distinct* elements; by homomorphism, they are pairwise-orthogonal idempotents. None can be zero: $\varphi(e_i) = 0$ would force $e_i = 0$ by injectivity, contradicting $e_i \neq 0$.

Hence $\mathbb{R}^k$ contains n pairwise-orthogonal nonzero idempotents. But by the same idempotent-counting argument applied to $(\mathbb{R}^k, +, \odot)$, the maximum is k . So $n \leq k$, contradicting $k < n$. $\square$

What the theorem rules out. The theorem says nothing about lossy approximation: it only forbids *exact* ring homomorphisms. In particular, neighbours of the zero map ($\varphi \approx 0$) trivially satisfy the relation $\varphi(x \odot y) \approx \varphi(x) \odot \varphi(y)$ since both sides are small, but they are not informative compressors. A useful codec must compress non-trivially while incurring a small *distributional* error on natural-spectrum products; that is the relaxation we use.

1.3 Why distributional preservation is enough

The codec is trained to minimise $\mathcal{L}_{\text{code}} = \text{MSE}(E(R) \odot E(L), E(R \odot L))$ over training distributions of reflectances R and illuminants L (Sec. 3.1). By the impossibility theorem this loss cannot be zero in general: it can only be small *in distribution*. The empirical question is whether the natural-spectra distribution is concentrated enough to make this small in practice.

Two pieces of evidence support the relaxation:

Effective dimensionality. Reflectance and illumination spectra over the visible band lie on a low-dimensional manifold; the top 6 principal components of our 1,013-spectrum training set capture more than 99% of the variance, and PCA-6, the principal component analysis (PCA) basis built from them, has Vora value $v=0.948$ relative to the CMF subspace (Sec. 4.3). Most of the $(\mathbb{R}^n, +, \odot)$ algebra is *never reached* by natural spectra; the codec only needs to be approximately multiplicative on the small region of the algebra that natural spectra and their products inhabit.

Empirical product accuracy. Across $B=1 \dots 10$ multi-bounce evaluations on our held-out test set, the proposed codec keeps the per-bounce ΔE_{94} below 0.05 for $B \geq 5$ (Sec. 3.2; multi-bounce columns of the hyperparameter-search records). Multi-bounce error is the most demanding evaluation of $\mathcal{L}_{\text{code}}$ because it iterates the encoder-decoder approximation multiple times in series; the modest empirical growth confirms that distributional preservation suffices for the rendering tasks considered.

2 Codec architecture details

Fig. 2 of the main paper gives the full pipeline at a glance; the remainder of this section zooms in on the codec itself.

The codec is a pair of small linear maps, $E : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $D : \mathbb{R}^k \rightarrow \mathbb{R}^n$, with $n=47$ wavelength bins (Sec. 3.3), whose weights are constrained non-negative through a softplus re-parameterisation. This section documents the parameterisation (Sec. 2.1), analyses the structure of the learned weights (Sec. 2.2), and characterises what the codec preserves under direct viewing (Sec. 2.3) and under repeated products (Sec. 2.4).

2.1 Softplus parameterisation

The encoder and decoder weights are stored as unconstrained tensors $\tilde{W}_{\text{enc}} \in \mathbb{R}^{k \times n}$ and $\tilde{W}_{\text{dec}} \in \mathbb{R}^{n \times k}$, and the non-negativity required by the rendering pipeline (Sec. 3.3 of the main paper) is enforced through a softplus reparameterisation,

$$W_{\text{enc}} = \frac{1}{\beta} \log(1 + \exp(\beta \tilde{W}_{\text{enc}})), \quad W_{\text{dec}} = \frac{1}{\beta} \log(1 + \exp(\beta \tilde{W}_{\text{dec}})), \quad (1)$$

with $\beta = 10$ (chosen empirically: smaller values yield non-sparse decoder columns, larger values introduce gradient spikes near the

origin). Initialisation: $\tilde{W} \sim 0.01 \cdot \mathcal{U}[0, 1]$, keeping the codec near $z \approx 0$ at start. Optimiser: Adam [Kingma and Ba 2014], learning rate 10^{-3} , default momenta, no weight decay.

2.2 Structure of the learned weights

This section records the structural properties of the learned weights (encoder rows, decoder columns, and the chromaticity hull spanned by the decoder) that matter for the basis-shape comparison in Sec. 4.2. The six decoder columns are smooth, band-localised spectra: column widths adapt to the spectral variability of the training set (narrow around 540 nm, broader in deep blue/red); several columns carry small anti-correlated side-lobes from the $\mathcal{L}_{\text{code}}$ multiplicativity term; and W_{enc} is *not* the transpose of W_{dec} (the encoder is a tighter band-pass analyser, the decoder a smoother reconstruction filter).

2.3 What the codec preserves

We do not claim that the decoded spectrum $\hat{s} = D(E(s))$ matches the input s pointwise on the wavelength axis. We claim the codec preserves what the spectrum *means* for rendering: its colour under viewing, and its colour under multiplicative composition with another spectrum (the operation a renderer evaluates at every shading point). This subsection quantifies both.

Setup. For each training reflectance R and each training illuminant L (corpora of Sec. 3.3; 1,013 + 759 spectra), we compare the CMF integral of the spectrum against the CMF integral of its codec round-trip. We report on the training distribution because it is the appropriate reference for what the codec preserves on the data it was fitted to; held-out test gives essentially identical histograms (within $0.05 \Delta E_{2000}$ mean), confirming the codec’s accuracy is structural rather than overfitting-driven. All differences are reported as ΔE_{2000} in the perceptual CIELAB space implied by the relevant white point. No rendering or light transport is involved: the codec is the only operation between input and output.

Four panels (Fig. 1):

- Reflectance round-trip under D65: $|\text{CMF}(R \cdot D_{65}) - \text{CMF}(\text{DER} \cdot D_{65})|$, normalised by the D65 white point.
- Illumination round-trip: $|\text{CMF}(L) - \text{CMF}(\text{DEL})|$, each illuminant Y -normalised against itself.
- Per-channel product round-trip: $|\text{CMF}(R \odot L) - \text{CMF}(D(\text{ER} \odot \text{EL}))|$ on 5,000 random (R, L) pairs.
- Self-product chain R^N for $N=1 \dots 10$ (a stylised proxy for N -bounce SSS where the same albedo is multiplied repeatedly): $|\text{CMF}(R^N \cdot D_{65}) - \text{CMF}(D(\text{ER})^N \cdot D_{65})|$ on 100 test reflectances.

Fidelity summary. Mean ΔE_{2000} : reflectance round-trip 3.32 (median 3.15, p95 5.72); illumination round-trip 1.79 (median 1.41, p95 4.07); $R \odot L$ product 1.66 (median 1.04, p95 5.09); R^{10} self-product median 0.003 (p95 5.37). The reflectance round-trip is the largest because the training set contains saturated reflectances (Munsell pure colours and synthetic sRGB-primary targets) that sit near the codec’s reconstruction limit; in actual scenes these spectra are encountered *combined* with illuminants, and panel (c) measures that combined error.

Why the product fares better than the reflectance alone. The codec is trained with $\mathcal{L}_{\text{e2e}}$ on the rendering quantity $R \odot L$ (Sec. 3.1), not

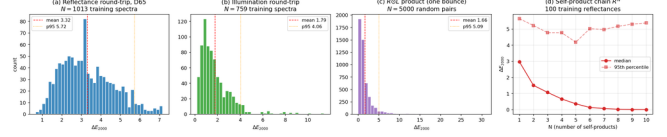


Fig. 1. Codec fidelity under viewing. Histograms of per-spectrum ΔE_{2000} for (a) reflectance round-trip, (b) illumination round-trip, (c) reflectance $\odot$ illumination one-bounce product (the rendering kernel), and (d) median + 95th-percentile ΔE_{2000} versus the number N of self-products R^N . None of these involve rendering through a transport simulator; the differences shown are purely codec error in the spectrum-as-seen-under-CMF sense.

on R in isolation. The training pressure is therefore strongest on the spectra and products that the renderer actually sees. Panel (c)’s median $\Delta E_{2000}=1.04$ on $R \odot L$ pairs sits below panel (a)’s 3.15 on reflectances alone, consistent with that asymmetry.

Why R^N decays under self-product. Reflectances $R \in [0, 1]^n$ shrink under repeated multiplication ($R^N \rightarrow 0$ as $N \rightarrow \infty$ wherever $R < 1$), and so does the codec round-trip; the perceptual gap converges to zero (panel (d): median 0.003 at $N=10$). The 95th-percentile stays near $5 \Delta E_{2000}$, contributed by the saturated training reflectances that approach 1 in some channels and 0 in others (these stay visible at large N). The graceful decay validates the codec’s $\mathcal{L}_{\text{code}}$ training under the iterated multiplications that arise in multi-bounce SSS: the codec does not amplify error with N .

2.4 Multi-bounce convergence (rendered)

The self-product check in panel (d) of Fig. 1 measures codec stability under repeated reflectance multiplication without invoking a renderer. Here we confirm that the same flatness carries through an actual path tracer at increasing bounce depth.

We render the Mitsuba matpreview scene (interior and exterior spheres on a diffuse plane) under two illuminants (broadband D65 and the CIE FL11 tri-band fluorescent) at five maximum-depth values $\{2, 3, 4, 5, 32\}$: direct only, one, two, three, and effectively unbounded bounces. The three reflectances and the FL11 narrowband illuminant are pre-screened against our codec so that the codec round-trip contribution to the rendered ΔE_{2000} stays below 1.4 on every cell; the ΔE_{2000} maps therefore visualise multi-bounce *convergence* rather than codec bias. Spectral ground truth is rendered with Mitsuba’s spectral build and CMF-integrated to sRGB; the latent path encodes the three reflectances and the illuminant through W_{enc} , runs two RGB passes through the unmodified RGB build, and decodes the pass outputs through W_{dec} . Per-pixel ΔE_{2000} versus spectral GT is whole-image, brightness-matched before computation; the colour bar uses uniform $v_{\text{max}}=5$.

Across both illuminants, codec error is essentially flat in bounce count. Under D65 the mean ΔE_{2000} moves from 0.90 (direct) to 0.91 (32 bounces), a 0.01-unit shift within rendering noise, and under FL11 from 1.30 to 1.32. The 95th-percentile is similarly flat (1.62 broadband, 1.93 FL11). Inter-reflections correctly fill in the shadowed regions (column “Ours” of Fig. 2) without amplifying codec error,

corroborating the conclusion of Sec. 2.3 under the actual rendering kernel rather than the stylised self-product.

3 Training details

Our codec is the outcome of a four-loss training objective (Sec. 3.1) selected by a two-phase 130-run hyperparameter search (Sec. 3.2) over the curated reflectance and illumination corpora described in Sec. 3.3. The per-component ablation (Tab. 1) confirms that each retained loss is necessary on at least one scene class.

3.1 Loss components: full formulas and rationale

The total objective is $\mathcal{L} = \lambda_{e2e}\mathcal{L}_{e2e} + \lambda_{rec}\mathcal{L}_{rec} + \lambda_{code}\mathcal{L}_{code} + \lambda_{col}\mathcal{L}_{col}$. Each term targets a distinct failure mode; the four together prevent any single one from being satisfied trivially.

End-to-end product loss $\mathcal{L}_{e2e}$.

$$\begin{aligned}\mathcal{L}_{e2e}(R, L) &= \Phi(D(E(R) \odot E(L)), R \odot L), \\ \Phi(x, y) &= (2 - \cos(x, y)) \text{MSE}(x, y),\end{aligned}\quad (2)$$

with $\cos(x, y)$ the cosine similarity along λ . This term trains the decoder on the radiance the renderer integrates. The $(2 - \cos)$ prefactor upweights MSE on shape mismatch, preventing a luminance-only fit.

Round-trip reconstruction $\mathcal{L}_{rec}$.

$$\mathcal{L}_{rec}(R, L) = \text{MSE}(\text{DER}, R) + \text{MSE}(\text{DEL}, L). \quad (3)$$

This term rules out the $E \equiv 0$ collapse that $\mathcal{L}_{e2e}$ alone admits, and pins the encoder behaviour on R and L individually rather than only on their product.

Latent multiplicativity $\mathcal{L}_{code}$.

$$\mathcal{L}_{code}(R, L) = \text{MSE}(E(R) \odot E(L), E(R \odot L)). \quad (4)$$

The relaxed homomorphism: the codec aligns latent products with the encoded interaction. It cannot be zero in general (Sec. 1), only small in distribution.

Colour $\mathcal{L}_{col}$.

$$\mathcal{L}_{col}(R, L) = \Delta E_{76}(\text{Lab}_{w(L)}(D(E(R) \odot E(L))), \text{Lab}_{w(L)}(R \odot L)), \quad (5)$$

under the sample-illuminant white point $w(L)$. This term provides perceptual gradient where the spectral losses are flat. We use ΔE_{76} rather than ΔE_{2000} during training (piecewise hue terms in ΔE_{2000} have irregular gradients); evaluation reports ΔE_{2000} .

A multi-bounce chain loss $\mathcal{L}_{mb}$ was also considered; the ablation (Sec. 3.2, Tab. 1) shows its work is subsumed by $\mathcal{L}_{code}$, so it is omitted from the final loss.

3.2 Hyperparameter search

The four loss weights were selected by a two-phase random search, following the protocol below.

Search protocol. The four weights $(\lambda_{rec}, \lambda_{e2e}, \lambda_{code}, \lambda_{col})$ were sampled log-uniformly from the ranges $\lambda_{rec} \in [0.01, 0.5]$, $\lambda_{e2e} \in [1, 8]$, $\lambda_{code} \in [0.1, 1]$, $\lambda_{col} \in [10^{-3}, 5 \times 10^{-2}]$, with bounds informed by a preliminary coarse search.

Phase 1. Eighty weight combinations were drawn independently and each codec was trained from scratch (300 epochs, batch size 256, learning rate 10^{-3} , early stop at patience 30, seed 777, single CPU process). Validation was on the stratified illuminant test split using the multi-bounce evaluation of ΔE_{94} across $B=1 \dots 10$ bounces; the run-level metric was the mean ΔE_{94} across B . The 10 best Phase-1 runs were additionally rendered on the four Cornell-box (cbox) illuminants to produce a render-time score.

Phase 2. The five Phase-1 runs with best composite rank (70% render score, 30% multi-bounce score) seeded a refinement search of 50 codecs. Each refinement sample drew weights uniformly within $\pm 30\%$ of the seed in log space, clamped to the global ranges above. Same training configuration and metrics as Phase 1.

Selected configuration. All $80 + 50 = 130$ trained codecs were ranked by the same multi-bounce ΔE_{94} metric augmented with the cbox render score on Phase-2 candidates; the selected configuration has weights

$$(\lambda_{rec}, \lambda_{e2e}, \lambda_{code}, \lambda_{col}) = (0.0111, 2.049, 0.939, 0.0232).$$

The weights span more than two orders of magnitude, reflecting the scale differences of their respective loss terms: $\mathcal{L}_{e2e}$ is per-wavelength MSE on $R \odot L \in [0, 1]^n$ values, while $\mathcal{L}_{col}$ is ΔE_{76} in CIELAB which ranges to several tens. Once the per-term scales are equalised by the weights, all four losses contribute commensurately to the total gradient.

Robustness. The top-5 Phase-2 configurations differ by $< 0.05 \Delta E_{94}$ across $B=1 \dots 10$: the loss landscape near the optimum is broad in weight space.

Ablation: which losses matter where. Re-training the codec with each loss weight set to zero or scaled by $1/10$ produces ten codecs in addition to the baseline. We evaluate each against the thirteen-cell scene suite and aggregate by scene class (Tab. 1).

Table 1. Ablation: mean ΔE_{2000} aggregated per scene class. “baseline” is our codec; each row is identical to baseline except for the named loss weight. Boldface marks ablations that worsen baseline by more than half its value on at least one scene class.

configuration	cbox	glass	conductor	SSS
baseline	1.69	1.56	1.28	3.05
$\lambda_{rec}=0$	1.94	1.57	1.10	6.52
$\lambda_{rec}/10$	1.76	1.27	1.10	3.73
$\lambda_{e2e}=0$	2.01	1.90	1.23	3.92
$\lambda_{e2e}/10$	1.91	1.74	1.57	4.78
$\lambda_{code}=0$	2.04	2.46	1.68	13.79
$\lambda_{code}/10$	1.83	1.92	1.46	6.40
$\lambda_{mb}=0$	1.70	1.63	1.28	3.57
$\lambda_{mb}/10$	1.70	1.62	1.28	3.52
$\lambda_{col}=0$	5.00	3.14	0.89	2.97
$\lambda_{col}/10$	1.80	1.55	1.34	2.40

The pattern is consistent with each loss’s design intent. $\mathcal{L}_{col}$ is essential for cbox and glass (the RGB-renderer scenes where direct colour fidelity drives the metric), and removing it triples cbox error. $\mathcal{L}_{code}$ is essential for SSS, where iterated self-products amplify

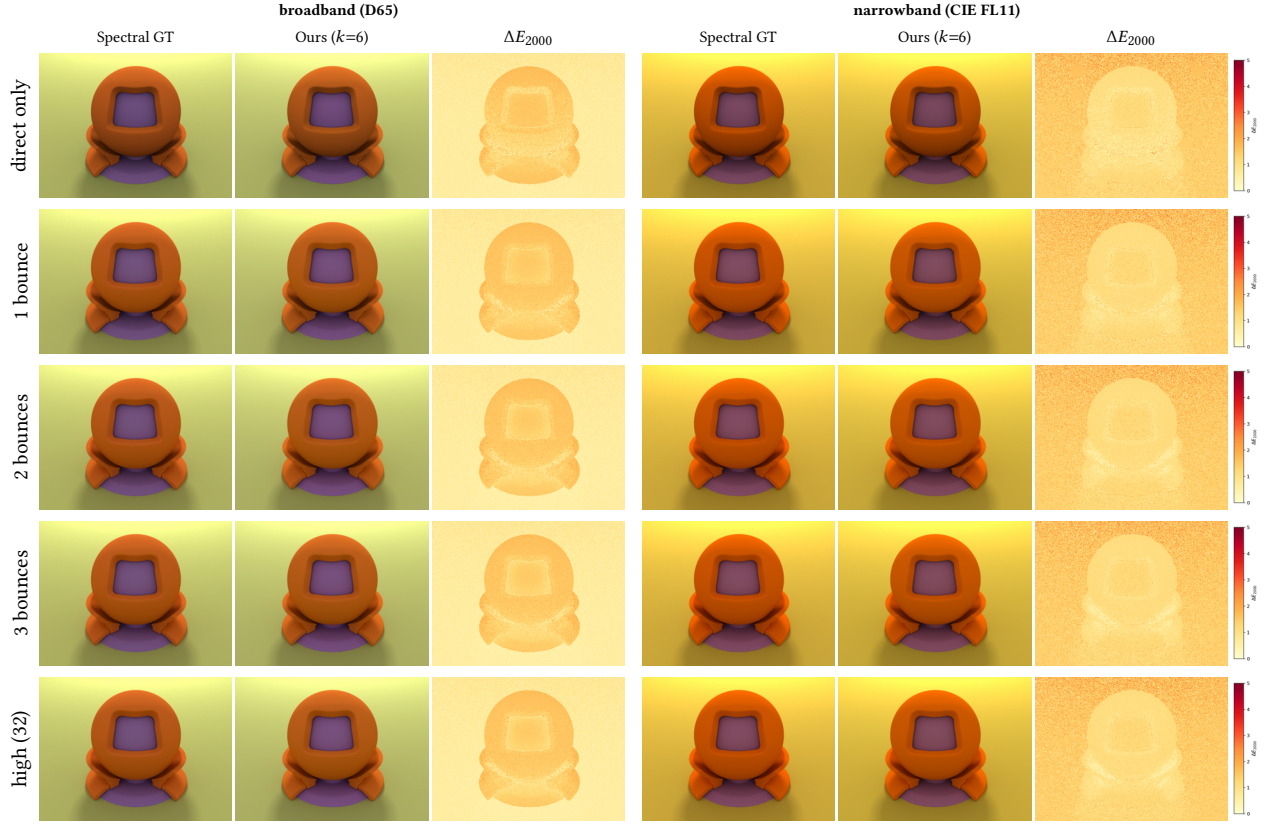


Fig. 2. **Multi-bounce convergence under broadband (D65, left) and narrowband (CIE FL11, right) illumination.** Maximum-depth values increase top to bottom: direct only (2), 1, 2, 3, and high (32) bounces. Columns per block: spectral GT (CMF-integrated to sRGB); our latent rendering at $k=6$ (two RGB passes through the unmodified RGB build); per-pixel ΔE_{2000} between the two (shared colour bar, right). Whole-image mean / p95 ΔE_{2000} : D65 0.90/1.62 (direct) through 0.91/1.62 (high); FL11 1.30/1.92 through 1.32/1.93, both flat to within rendering noise. FL11 is markedly narrower in SPD and warmer in chromaticity than D65, giving the scene a visibly different tone; the small per-cell offset over the D65 numbers is the codec’s narrowband round-trip floor, not bounce-accumulated error.

any multiplicativity gap; removing it grows SSS error from 3.05 to 13.79 ΔE_{2000} . $\mathcal{L}_{mb}$, the multi-bounce chain loss considered during development, shifts every scene class by $< 0.6 \Delta E_{2000}$, indicating that its work is already covered by $\mathcal{L}_{code}$. Our codec therefore uses the four losses $\{\mathcal{L}_{e2e}, \mathcal{L}_{rec}, \mathcal{L}_{code}, \mathcal{L}_{col}\}$.

3.3 Training data: corpus construction and splits

The codec is trained on disjoint reflectance and illumination corpora. Both are sampled to the 47-channel grid (368–830 nm uniformly spaced, matching the spectral Mitsuba 0.6 build used for ground truth) and zeroed outside 400–700 nm, the range covered by the measured reflectance data.

Reflectance corpus. 1,449 spectra in three groups: (i) 1,269 measured Munsell-system reflectances; (ii) 36 *optimal* reflectances targeting sRGB primaries via bounded least-squares (saturation $s \in [0.6, 0.98]$, luminance $Y=0.30 Y_{max}$): for each primary $c_t=(X_t, Y_t, Z_t)$ we solve $\min_s \|w_X^T s - X_t\|^2 + \|w_Y^T s - Y_t\|^2 + \|w_Z^T s - Z_t\|^2$ for the 47-channel reflectance $s \in [0, 1]^{47}$ under D65; (iii) 144 synthetic smooth

reflectances by basis-function optimisation (24 hue directions $\times$ 6 saturation levels, $s \in [0.7, 0.98]$). The *optimal* and synthetic-smooth groups extend coverage toward saturated colours under-represented in measured datasets and into the sRGB-edge region. Split: 70/30 train/test using *adaptive ring-sector partitioning* in CIELAB a^*b^* : 180 angular bins $\times$ adaptive radial quantiles, with 70% of each cell randomly assigned to training. This avoids spatial clustering and ensures both splits span the full hue/saturation range. Result: **1,013 train / 436 test**.

Illumination corpus. Sources combined: LSPDD-measured lamp spectra [LSPDD 2026] (compact fluorescent, halogen, LED, metal halide, incandescent, CIE standard illuminants), simulated CIE day-light phases, 82 synthetic broadband spectra varying correlated colour temperature (CCT), 367 narrowband spectra with sharp emission peaks, and vertically flipped variants of the broadband spectra (to enrich shape diversity). *Resampling preserves narrow features.* LSPDD lamp spectra are re-sampled to the 47-channel grid via *bin integration* (sum of spectral power within each bin’s wavelength range), not point sampling. Bin integration is essential because point

sampling at the ~ 10 nm grid would alias the narrow emission peaks of fluorescent lamps and similar sources, distorting their colour and energy. *Greedy similarity deduplication.* After resampling and unit-peak normalisation we apply iterative cosine-similarity deduplication at $\tau=0.95$, removing near-duplicate shapes. This yields **708 curated spectra**. Split via 36 angular bins: **497 train / 106 val / 105 test**.

Augmentation, val/test mirroring. A separate augmentation pass enriches the training set and the narrowband content of val/test. Training adds 262 synthetic spectra: 100 convex combinations of LSPDD pairs ($\alpha \in [0.2, 0.8]$), 81 jittered copies ($\pm 15\%$ multiplicative noise + sub-bin peak shifts via interpolation), and 81 broadband mixes. Val and test each gain 27 measured LSPDD spectra, disjoint from training, so that early stopping monitors narrowband accuracy. Final dataset: **759 train / 133 val / 132 test**.

4 Capacity analysis (full)

The main paper claims that $k=6$ is the operating point that makes the codec useful in practice. This section supplies the empirical evidence: $k=3$ cannot simultaneously satisfy CMF-subspace alignment and per-channel-product preservation (Sec. 4.1); under matched encoder training, no fixed-basis $k=6$ alternative (PCA, NMF, box, or fitted-Gaussian) matches our codec on the rendered metric (Sec. 4.2); subspace alignment to the CMFs is necessary but not sufficient (Sec. 4.3); and a fresh $k=9$ codec yields a negligible aggregate gain over the $k=6$ codec across thirteen scenes (Sec. 4.4), so the third RGB pass is unwarranted.

4.1 The $k=3$ Pareto trade-off

No learned $k=3$ codec we trained simultaneously satisfies (i) direct-illumination accuracy (Luther alignment) and (ii) multi-bounce-product preservation; together with the impossibility theorem, the empirical Pareto frontier rules out the corner where both would hold. The two paragraphs below give the evidence for each architectural family (non-negative and signed W_{dec}) that we evaluated.

Non-negative decoder. A 130-run two-phase search (same protocol as Sec. 3.2) on $k=3$ codecs with non-negative W_{dec} converges to a configuration whose Vora value is $\nu=0.691$ (max principal angle 46° , Sec. 4.3), well short of the Luther target $\nu=1$. On the four-illuminant cbox suite of the search, this codec renders at mean $\Delta E_{2000} \approx 5.03$, well above the RGB baseline (Tab. 3). In other words, the best non-negative learned $k=3$ codec is *worse* than RGB on direct colorimetry, and offers no compensating multi-bounce advantage.

Signed decoder. Standard RGB pipelines tolerate a signed colour-conversion matrix (sRGB $\rightarrow$ XYZ has negative entries). Allowing signed weights in W_{dec} (while keeping the encoder non-negative as required by the renderer's per-channel architecture) shifts the Pareto frontier: a sign-relaxed counterpart of the same search reaches $\nu=0.821$, a single-bounce direct-only $\Delta E_{2000}=1.42$, below RGB's 2.09, but a full-cbox $\Delta E_{2000}=2.93$ that is slightly *worse* than RGB's 2.68. The composite (both direct and indirect averaged) is 2.17 vs. 2.39, a marginal 9% improvement.

Why this is a Pareto trade-off, not just a tuning failure. The non-negative-decoder family and the signed-decoder family *both* fall on (different points of) a frontier in the (ν , product-preservation error) plane: lowering product error (by training $\mathcal{L}_{\text{code}}$ harder) pulls ν further from 1, and vice versa. No single $k=3$ codec sits at the corner ($\nu=1$, product error=0); the impossibility theorem (Sec. 1) says no such codec can exist for $k < n$. Increasing capacity to $k=6$ removes the trade-off at the operating points we tested: our codec achieves $\Delta E_{2000}=1.45$ on the four-illuminant cbox suite of the search at $\nu=0.74$, better on the rendered metric than *either* $k=3$ corner.

4.2 Capacity baselines: does learning help, and does the basis shape matter?

A learned six-dimensional spectral basis could in principle be replaced by a hand-designed alternative: six contiguous box bins, six smooth Gaussians, or a data-derived basis from PCA or non-negative matrix factorisation. We isolate two factors. First, *does encoder training help*: holding the decoder fixed, what does training the encoder under our loss configuration buy? Second, *does the basis shape itself matter*: with the encoder fully trained, does the choice of decoder (box, Gaussian, or learned) change the result? Both factors matter, they are not interchangeable, and no fixed-basis variant we evaluated matches the learned codec.

We compare seven configurations at $k=6$ against the RGB baseline (effectively $k=3$). Two **data-fitted** decoders (PCA-6 from the top-six right singular vectors of our training spectra; NMF-6 from non-negative matrix factorisation on the same data) are evaluated with their natural inference rule (a least-squares projector for PCA; a clipped projector for NMF). Two **hand-crafted** decoder shapes (six contiguous 50 nm boxes; six Gaussians fitted by least squares to our codec's decoder columns) appear in two forms each: a *handcrafted* form with $W_{\text{enc}} = \text{clip}(\text{pinv}(W_{\text{dec}}), 0)$ and no training, and a *trained* form where the encoder is trained from scratch under our exact loss configuration with the decoder frozen. The **learned** entry is our $k=6$ codec with both W_{enc} and W_{dec} trained end-to-end. The trained-encoder Box and Gaussian variants share the same training data, loss weights, and 300-epoch schedule; the only variable is the decoder shape.

The seven configurations are:

- **PCA-6**: data-fitted W_{dec} (signed) and natural least-squares projector encoder; no training. Latent codes are clipped to ≥ 0 at encode time (the renderer cannot accept negative material values), which destroys most of the encoded information when projected spectra produce negative codes.
- **NMF-6**: data-fitted W_{dec} (non-negative by construction) and least-squares projector encoder; no training. Reconstructs spectra well in a Frobenius sense but is not trained for the rendering pipeline's multiplicative composition.
- **Box-6 handcrafted**: W_{dec} = six contiguous equal-width 50 nm bins covering 400–700 nm; encoder set to the clipped pseudoinverse; no training.
- **Box-6 trained**: same Box W_{dec} , encoder trained under our codec's loss with the decoder frozen.
- **Gauss-fit-6 handcrafted**: W_{dec} = six Gaussian kernels fitted by least squares to our codec's decoder columns (closest

single-Gaussian approximation of our learned basis); encoder set to the clipped pseudoinverse; no training.

- **Gauss-fit-6 trained:** same Gauss W_{dec} , encoder trained under the same loss as our codec.
- **Ours**, $k=6$: both W_{enc} and W_{dec} trained end-to-end.

Does encoder training help? The cleanest answer comes from the Box and Gaussian variants, which appear in both untrained and encoder-trained forms. Tab. 2 reports Cornell-box five-illuminant mean ΔE_{2000} .

Table 2. Effect of training the encoder under our codec’s loss with the decoder frozen. Cbox mean ΔE_{2000} over the five illuminants of Tab. 1 of the main paper.

basis	no training	encoder trained	drop
Box-6	4.46	2.36	−47%
Gauss-fit-6	14.46	2.57	−82%

Encoder training reduces error by a factor of roughly 2 to 6. Both trained variants end below the RGB baseline ($\Delta E_{2000}=3.31$ on the same suite). The PCA-6 and NMF-6 entries are not directly comparable because their encoders are not trained under our losses, but even so, neither trained variant matches our codec at $\Delta E_{2000}=1.24$, so the basis shape itself still nearly doubles the error on top of the encoder-training gain.

Does the basis shape matter? Yes, and the per-illuminant breakdown (Tab. 3) is the clearest place to see how.

Table 3. Cornell-box per-illuminant mean ΔE_{2000} : Tab. 1 of the main paper extended with the untrained bases (“hc”: handcrafted encoder, no training). Best per illuminant in bold. All rows are at 1024 spp, as in Tab. 1 of the main paper, except the handcrafted (hc), NMF-6 and PCA-6 rows, which keep the 256 spp budget of the per-cell tables of Sec. 4.5 outside the narrow₅₂₅ column.

basis	broad ₁₂	broad ₆₃	narrow ₅₂₅	narrow ₃₅₁	FL11	avg
Ours , $k=6$	0.90	1.52	1.11	0.86	1.83	1.24
Box-6 trained	1.25	2.03	2.94	1.42	4.14	2.36
RGB baseline	3.40	2.42	2.89	3.82	4.01	3.31
Gauss-fit-6 trained	0.92	1.81	1.50	1.10	7.52	2.57
Box-6 hc	2.15	2.40	1.87	10.47	5.39	4.46
NMF-6	8.70	11.01	8.40	8.10	14.98	10.24
Gauss-fit-6 hc	11.47	15.83	14.79	20.93	9.27	14.46
PCA-6	40.70	43.13	47.69	46.48	38.95	43.39

The trained Box and Gauss-fit variants are competitive with our codec on the smooth broadband illuminants but each degrades on a different narrowband cell. Box-trained loses primarily on FL11 (4.14 vs. our codec’s 1.83): the discrete bin-and-average operation cannot resolve FL11’s three narrow emission peaks. Gauss-fit-trained loses dramatically on FL11 (7.52): the narrow Gaussians fitted to our codec’s columns miss the cracks between FL11’s peaks more severely than Box’s bins do. Our codec has the lowest ΔE_{2000} on every illuminant in the suite. The FL11 collapse on Gauss-fit-trained is the sharpest visual evidence that fitting Gaussians to our learned basis produces a *worse* basis than ours, even though the Gaussians

are by construction the closest single-Gaussian approximation of our codec’s columns.

Fig. 5 of the main paper shows the visual artefact modes of each basis on the FL11 cell, where the differences are most pronounced.

On the Vora value (Luther condition). The Vora value ν measures how well the decoder subspace aligns with the CIE colour-matching functions [Vora and Trussell 1993] and is sometimes proposed as a metric of basis quality. On this ladder it is not predictive of rendering quality (Tab. 4).

Table 4. Vora value ν (CMF subspace alignment, $0 \leq \nu \leq 1$), maximum principal angle to span(CMF), and rendered cbox mean ΔE_{2000} (five-illuminant average; “hc”: handcrafted encoder, no training). The highest- ν decoder is also the worst render.

basis	ν	max-angle	cbox ΔE_{2000}
PCA-6	0.95	19.2°	43.39
NMF-6	0.94	18.8°	10.24
Box-6 trained / hc	0.83	32.4°	2.36 / 4.46
Gauss-fit-6 trained / hc	0.79	32.7°	2.57 / 14.46
Ours , $k=6$	0.74	36.2°	1.24

PCA-6 has the highest ν because its top-six singular vectors span the maximum-variance directions of the training data, including the smooth components that overlap with the CMFs. But its signed entries cause about two thirds of the latent codes on test spectra to go negative, and clipping at the renderer destroys most of the encoded information. NMF-6 is also high- ν , non-negative by construction, and reconstructs spectra well, but its encoder is not trained under any rendering loss, and its basis was not trained for the multiplicative composition that arises in our two-pass rendering pipeline. Conversely, our codec has the *lowest* ν of the seven $k=6$ variants and the best rendering quality.

We do not claim a direct causal link between the codec’s lower ν and its better rendering. The empirical pattern is simpler: a basis must satisfy three concurrent requirements: the codes must be non-negative (the renderer treats them as material values), the basis must span enough of the CMF subspace for colorimetric accuracy, and the encoder and decoder must be jointly adapted to the multi-bounce, multi-light products that the rendering pipeline computes. Vora measures only the second. We therefore use ν here as an *indicator* (it identifies which failure mode a basis exhibits) rather than as the metric of basis quality.

4.3 Vora value: definition and full codec table

The classical *Luther condition* requires that the trichromat basis (in our case, $W_{\text{dec}} \in \mathbb{R}^{n \times k}$) span the same subspace as the CIE colour-matching functions $\text{CMF}^T \in \mathbb{R}^{n \times 3}$. Vora and Trussell [1993] measure the degree of subspace alignment by

$$\nu = \frac{1}{3} \text{tr}(P_B P_A) = \frac{1}{3} \sum_{i=1}^3 \cos^2 \theta_i, \quad (6)$$

where P_A, P_B are the orthogonal projectors onto span(CMF^T) and span(W_{dec}) and θ_i are the principal angles. $\nu=1$ means Luther’s condition is satisfied exactly; $\nu=0$ means the subspaces are orthogonal.

Computation: orthonormalise both via QR, take singular values of the inner-product matrix.

Tab. 5 extends the 5-codec table of Sec. 4.2 with the $k=3$ codecs. The ordering by descending ν confirms the point made earlier: high ν alone does not guarantee low rendering error: PCA-6 ($\nu=0.948$) renders catastrophically, NMF-6 ($\nu=0.942$) is worse than RGB, while our $k=6$ codec has the lowest ν of any non-negative $k=6$ trained codec yet achieves the lowest rendering ΔE_{2000} . Within a fixed architecture the Spearman correlation between $\theta_{\max}$ and rendered ΔE_{2000} is only $\rho \approx 0.59$ (over the 16 rendered search codecs).

Table 5. Vora value ν and maximum principal angle $\theta_{\max}$ for all codecs evaluated in this document.

Codec	k	ν	$\theta_{\max}$
RGB baseline ($W_{\text{dec}} = \text{CMF}^T$)	3	1.000	0.0°
PCA-6 (top-6 PCs of training reflectances)	6	0.948	19.2°
NMF-6	6	0.942	18.8°
Box-6 (handcrafted equal-width bins)	6	0.833	32.4°
Signed W_{dec} at $k=3$	3	0.821	39.8°
Gauss-fit-6 (fit to our decoder)	6	0.788	32.6°
Ours, $k=6$ (final codec)	6	0.736	36.2°
Best non-neg $k=3$ codec	3	0.691	45.7°
$\mathcal{L}_{\text{col}}$ -only $k=3$	3	0.300	66.7°

A $k=3$ codec trained with $\mathcal{L}_{\text{col}}$ alone converges to $\nu=0.30$, well below the best joint-loss $k=3$ codecs ($\nu \approx 0.6-0.69$; search median 0.52). The colour loss is a perceptual-difference penalty on natural reflectances, not a pressure to span the CMF subspace, so it is not a Luther proxy.

4.4 $k=9$: more capacity does not help

The main paper claims $k=6$ is the practical operating point: $k=3$ loses to RGB on the multi-bounce side of the Pareto frontier, while $k=9$ requires three RGB passes for no commensurate quality gain. Here we give the empirical evidence for the $k=9$ side of that claim.

We trained a fresh $k=9$ codec following the same protocol as our $k=6$ codec (4-loss configuration, 130-run hyperparameter search, same training data of Sec. 3.3), then evaluated the best run on the same thirteen scene/illuminant cells used to evaluate our $k=6$ codec across material classes (Tab. 6).

The aggregate gain is negligible, and the per-cell differences trade in both directions rather than favouring $k=9$ systematically (Tab. 6).

The saturation is consistent with two properties established elsewhere in this document. First, the data manifold is nearly exhausted: six principal components capture 99.3% of the reflectance-corpus variance (Sec. 1.3), and the additional channels mainly refine illumination detail (85.9% at $k=6$ rising to 93.8% at $k=9$). Second, the reported error is colorimetric: rendered spectra are integrated against three colour-matching functions before comparison, so added spectral resolution contributes increasingly along metameric directions that the integration cannot register. The error remaining at $k=6$ is instead dominated by the product approximation, which the impossibility theorem ties to any finite k , and by sources concentrated between basis functions; neither improves at $k=9$, where narrow₃₅₁ regresses while the aggregate stays flat.

Table 6. Per-scene mean ΔE_{2000} , $k=9$ vs. $k=6$. Aggregate $\Delta \approx -0.05$ across thirteen scenes; $k=9$ has the lower ΔE_{2000} on six of the thirteen rows (by more than 0.1 on four: Ag, Cu, SSS skin, and SSS wholemilk), loses on narrow₃₅₁ by 0.6 and on SSS potato by 0.36, and trades within 0.3 elsewhere.

scene/light	$k=6$	$k=9$	Δ
cbox broad ₁₂	1.08	1.30	+0.22
cbox broad ₆₃	1.68	1.61	-0.07
cbox narrow ₃₅₁	1.10	1.70	+0.60
cbox FL11	1.92	1.86	-0.06
glass red (transmit)	1.16	1.45	+0.29
glass blue	1.44	1.45	+0.01
glass amber	1.71	1.74	+0.03
conductor Au	0.91	0.92	+0.01
conductor Ag	1.19	0.86	-0.33
conductor Cu	1.02	0.82	-0.20
SSS skin	2.63	2.35	-0.28
SSS potato	2.66	3.02	+0.36
SSS wholemilk	3.85	2.62	-1.23
aggregate	1.72	1.67	-0.05

Against this we must weigh the cost: $k=9$ requires three RGB passes per render instead of two, a 50% increase in rendering time at fixed samples per pixel (spp). The codec is preferred at $k=6$. Fig. 3 shows the thirteen evaluation cells.

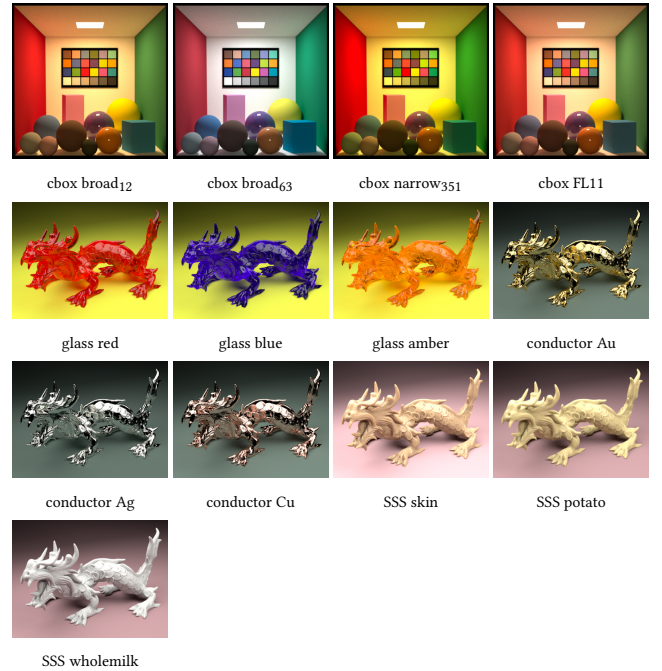


Fig. 3. The thirteen scene/illuminant cells of the $k=9$ vs. $k=6$ sweep (spectral-GT renders): the four Cornell-box illuminants, three coloured glasses, three measured conductors, and three subsurface materials.

4.5 Per-cell ΔE_{2000} tables

Tab. 7 reports the per-cell ΔE_{2000} for each scene’s four illuminants (broad₁₂ / broad₆₃ / narrow₃₅₁ / FL11 for cbox; D65 / narrow₄₇₅ / narrow₅₂₅ / narrow₆₂₅ for the materials); each scene block closes with its four-light mean.

Two patterns emerge. First, no fixed-basis configuration matches our codec across all five scenes: Box-6 handcrafted has the lowest ΔE_{2000} on conductor and SSS skin but loses by 3.5 $\times$ on cbox. Second, training the encoder on top of a hand-picked decoder closes most of the gap: encoder-trained Box-6 trails our codec by only 0.3–0.5 ΔE_{2000} on glass and conductor, but degrades catastrophically on SSS skin (8.5 vs. 2.8). Our codec is the only configuration that delivers low error on every scene class without per-scene retuning.

Table 7. Per-cell mean ΔE_{2000} for all 5 codecs, 5 scenes, 4 lights = 100 evaluation cells; each scene block closes with its four-light mean (bold: best per row). “hc” = handcrafted (frozen encoder); the unlabelled “Box-6” / “Gauss-fit-6” rows train W_{enc} on the corresponding frozen W_{dec} .

Scene	Light	Ours	Box-6	Gauss-fit-6	Box-6 hc	Gauss-fit-6 hc
cbox	broad ₁₂	1.08	1.42	1.09	2.15	11.47
	broad ₆₃	1.68	2.21	1.92	2.40	15.83
	narrow ₃₅₁	1.10	1.63	1.26	10.47	20.93
	FL11	1.92	4.19	7.53	5.39	9.27
	mean	1.45	2.37	2.95	5.11	14.38
conductor (Cu)	D65	1.01	1.19	1.31	1.95	10.85
	narrow ₄₇₅	5.00	5.38	5.78	1.05	22.13
	narrow ₅₂₅	0.97	1.72	0.98	1.43	9.60
	narrow ₆₂₅	1.05	1.16	1.48	0.74	0.89
	mean	2.01	2.36	2.39	1.29	10.87
glass-red (transmit)	D65	1.34	2.80	2.42	3.02	2.68
	narrow ₄₇₅	6.05	5.69	5.78	4.99	12.02
	narrow ₅₂₅	1.82	2.06	1.51	2.89	5.66
	narrow ₆₂₅	1.36	1.21	2.40	0.86	3.56
	mean	2.64	2.94	3.03	2.94	5.98
glass-red (medium)	D65	2.41	1.46	2.27	3.89	4.71
	narrow ₄₇₅	5.87	6.41	5.15	5.10	20.87
	narrow ₅₂₅	1.52	1.45	1.03	1.23	10.37
	narrow ₆₂₅	1.51	3.27	1.25	1.02	1.49
	mean	2.83	3.15	2.43	2.81	9.36
SSS skin	D65	3.00	7.91	4.40	2.93	18.43
	narrow ₄₇₅	3.74	10.68	6.18	3.00	45.53
	narrow ₅₂₅	1.65	5.71	2.59	2.55	26.23
	narrow ₆₂₅	2.80	9.62	1.99	1.71	5.61
	mean	2.80	8.48	3.79	2.55	23.95

Two patterns are visible at the per-cell level. Narrowband-625 nm favours the hand-picked Box-6 basis on all four scenes that contain that cell (conductor, glass transmit, glass medium, SSS skin): Box-6’s sixth bin sits exactly on the deep-red sub-band and is well matched to a 625 nm illuminant. SSS skin is the most sensitive scene; skin’s spectrally rich reflectance interacts with multi-bounce subsurface transport in a way that exposes decoder spectral fidelity. Here the handcrafted Box-6 basis and our codec are close (2.55 vs. 2.80 on average), while the encoder-trained Box-6 and both Gaussian variants degrade sharply (8.48, 3.79, 23.95). Fig. 5 of the main paper shows the corresponding rendered comparison on the FL11 exemplar cell; all remaining cells use the identical rendering and visualisation pipeline.

5 Robustness under narrowband illumination

The cbox suite of Sec. 4.2 contains exactly one structured red-dominant source (narrow₃₅₁, main peak 670 nm with secondary lobes near 540 and 410 nm; the 525 nm Gaussian column of Tab. 1 of the main paper is a cell of the sweep below) and one multi-peaked source (FL11). This is enough to motivate the codec but not enough to characterise its narrowband behaviour. We therefore sweep the codec across a 52-condition grid of Gaussian narrowband illuminants spanning the visible band and four bandwidths, on the same Cornell-box scene as the cbox suite.

Setup. Centres $\in \{400, 425, \dots, 700\}$ nm (13 values, 25 nm step); bandwidths FWHM $\in \{40, 60, 80, 100\}$ nm (4 values). For each (centre, FWHM) cell we render three images: spectral GT (Mitsuba 0.6 spectral build), RGB baseline (Mitsuba 0.6 RGB build with CMF-collapsed reflectances and illuminant), and our latent two-pass codec result. We report mean and 95th-percentile ΔE_{2000} over the rendered image; both error metrics are computed against the spectral GT.

Headline. Across all 52 cells, the codec averages mean $\Delta E_{2000}=2.12$ versus 19.4 for the RGB baseline; averaged per cell, the codec’s advantage is 11.2 $\times$ (median 10.6 $\times$); Tab. 8 breaks the result down by bandwidth. The codec’s worst cell, 475 nm at FWHM 40 nm with $\Delta E_{2000}=5.52$, is still better than the RGB baseline’s best cell, 450 nm at FWHM 100 nm with $\Delta E_{2000}=11.4$, so the codec’s advantage holds in every cell of the suite.

Table 8. Mean ΔE_{2000} marginalised over illuminant centre, by bandwidth. Wider bandwidths are easier for the codec; the worst FWHM (40 nm) is also where the codec’s spectral-basis structure is most exposed. RGB baseline error is roughly constant across bandwidth (a CMF-collapsed renderer cannot reconstruct narrowband features regardless of width).

FWHM (nm)	latent (ours)	RGB baseline	ratio
40	2.71	19.96	7.4 $\times$
60	1.97	19.46	9.9 $\times$
80	1.93	19.27	10.0 $\times$
100	1.88	18.98	10.1 $\times$

The full 13 $\times$ 4 grid is shown in Fig. 4.

Two basis-gap peaks. At the narrowest bandwidth (FWHM 40 nm), per-centre mean ΔE_{2000} shows two pronounced peaks (Tab. 9) at 475 nm ($\Delta E_{2000}=5.52$) and 575 nm ($\Delta E_{2000}=4.77$), while ~ 525 nm and ~ 625 nm sit at $\sim 1.1 \Delta E_{2000}$.

The peak at 475 nm sits between our codec’s decoder columns peaked near 438 nm and 509 nm; the peak at 575 nm sits between columns peaked near 539 nm and 619 nm (Sec. 2.2). When a narrow-band emission line falls in such a gap, no single decoder column has strong support there, and the encoder is forced to express the illuminant as a roughly equal mixture of the two flanking columns. The decoded result is a colour mixture rather than a clean narrow-band. The peaks at 400 nm and 700 nm are at the visible-band edge where neither training data nor CMF response is strong.

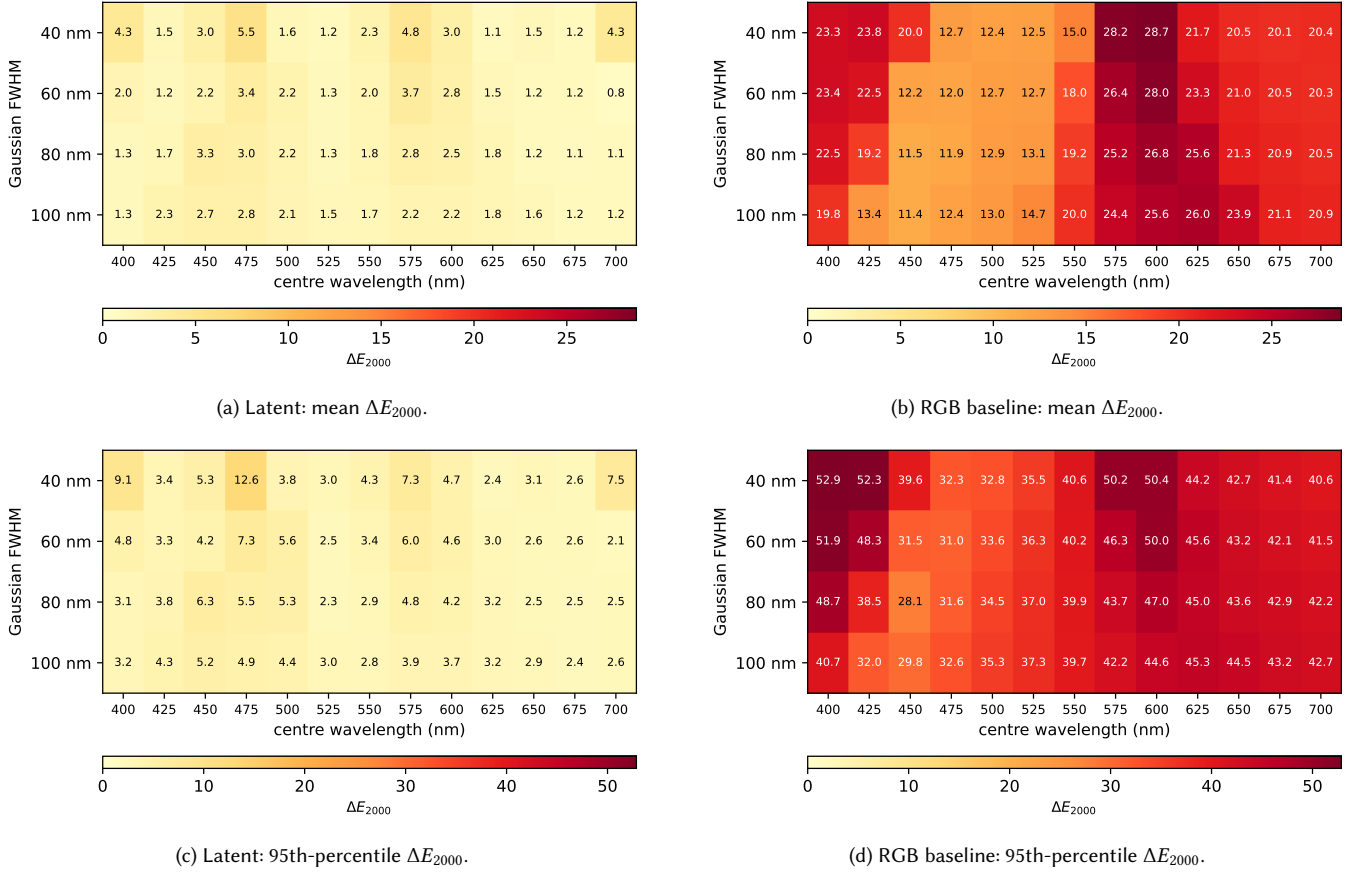


Fig. 4. Codec evaluation on the Cornell box under a narrowband Gaussian illuminant scan. Top row: image-mean ΔE_{2000} for our codec (a) and the RGB baseline (b). Bottom row: 95th percentile. Latent (left) and RGB (right) share a common colour scale per row, so the cross-method gap is directly readable from the colour intensity rather than the numeric labels.

Table 9. Mean ΔE_{2000} at FWHM 40 nm, scanning the illuminant centre. The peaks at 475 and 575 nm sit between our codec’s decoder columns; the valleys at 525, 625 and 675 nm sit on column peaks.

centre (nm)	latent (ours)	RGB baseline	centre (nm)	latent (ours)	RGB baseline
400	4.27	23.3	575	4.77	28.2
425	1.50	23.8	600	2.96	28.7
450	2.99	20.0	625	1.08	21.7
475	5.52	12.7	650	1.54	20.6
500	1.63	12.4	675	1.16	20.1
525	1.17	12.5	700	4.28	20.4
550	2.30	15.0			

Visual artefact mode. At the basis-gap centres, the artefact is a hue shift toward the mean of the two neighbouring decoder columns, not a luminance error. At 475 nm specifically, the codec’s blue narrowband renders as violet because the illuminant’s energy is split between the 438 nm and 509 nm columns. Fig. 5 shows the rendered images and per-pixel error maps at the two worst narrowband cells.

The codec error is hue-shifted but spatially diffuse; the RGB baseline collapses the entire scene into a chromaticity that cannot be reconstructed from a single broadband exposure.

Limit of the codec at $k=6$. The basis-gap mode is exactly what the impossibility theorem (Sec. 1) predicts at $k < n$: there are spectra (here, narrowband peaks centred in basis gaps) for which the codec’s reconstruction loses information that no amount of training recovers. Increasing k to 9 or higher would shrink the gaps but adds rendering passes; the trade-off is the central capacity argument of Sec. 4.4. Within the practical $k=6$ regime, narrowband sources offset by 25 nm or more from a basis-gap centre are within 1–2 ΔE_{2000} of the spectral ground truth.

6 Material encoding

The codec is content-agnostic: any wavelength-indexed quantity $f(\lambda) \in \mathbb{R}^{47}$ encodes through the same $E: \mathbb{R}^{47} \rightarrow \mathbb{R}^k$ regardless of its physical meaning. This section documents how the spectral inputs of the four material classes used in the experiments (coloured glass, conductors, and homogeneous / heterogeneous participating media)

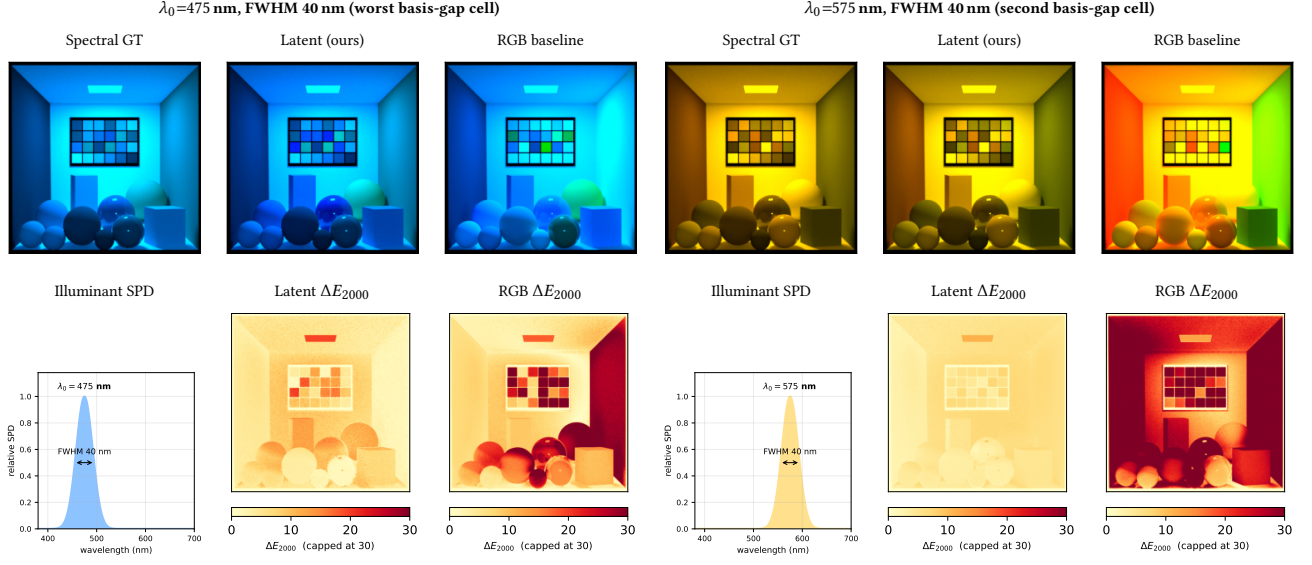


Fig. 5. **Narrowband-illumination exemplars at the codec’s two basis-gap cells**, $\lambda_0=475$ nm (left) and 575 nm (right), both Gaussian with FWHM 40 nm. Top row: spectral GT, latent (ours), RGB baseline; bottom row: illuminant SPD and the two error maps (shared colour scale, cap 30). At 475 nm the latent render shows a uniform hue shift toward violet but preserves spatial structure (mean / p95 ΔE_{2000} 5.52/12.60), while the RGB baseline collapses the scene into a near-monochromatic blue (12.72/32.28, ~ 33 on saturated reflectances). At 575 nm the mode is the same: latent 4.77/7.26 against RGB 28.24/50.23.

are routed through the codec into the corresponding unmodified Mitsuba plugin.

Coloured glass. Coloured glass is characterised by a spectral transmittance $\tau(\lambda) \in [0, 1]$: the wavelength-dependent fraction of incident radiance that passes through the material. Depending on the optical path through the coloured medium, that transmittance enters light transport in one of two distinct ways. We encode it once into a latent code $z_\tau = E(\tau) \in \mathbb{R}^k$ and pass it through the appropriate regime.

Surface-transmittance regime (thin filters). When the path through coloured material is short relative to a reference depth (a thin filter, a single dielectric crossing in a hollow vessel), the transmittance attenuates radiance at the surface: $L_{\text{out}}(\lambda) = \tau(\lambda) L_{\text{in}}(\lambda)$. Across N surface crossings the cumulative attenuation is $\tau(\lambda)^N$, which in the latent representation is the integer Hadamard power z_τ^N . This is exactly the algebraic structure the codec was trained to preserve (Sec. 3), so repeated crossings compound only the codec’s product error, which stays below $5 \Delta E_{2000}$ through twelve crossings (Fig. 8a).

Beer–Lambert volumetric-absorption regime (bulk glass). For solid coloured volumes (a 1 cm-thick cast dragon, a glass paperweight), the spectral attenuation depends on the optical path length d through the absorbing medium: Beer’s law gives $T(d, \lambda) = \exp(-\sigma_a(\lambda) d)$. We obtain a latent absorption coefficient by inverting the same transmittance code at a reference depth d_{ref} ,

$$\sigma_a^{(p)} = -\ln(z_\tau^{(p)}) / d_{\text{ref}},$$

with $z_\tau^{(p)}$ clipped to $[10^{-6}, 1]$ before the inverse so that $\sigma_a^{(p)}$ stays finite and non-negative. The volume integrator accumulates per-channel transmittance $\exp(-\sigma_a^{(p)} d)$ along each ray segment, which is a *fractional* Hadamard power of the encoded τ : the codec preserves it approximately rather than exactly, and the end-to-end loss $\mathcal{L}_{e2e}$ is what bounds the resulting error in practice. d_{ref} is set to the median path length through the coloured medium in each scene (~ 1 cm solid dragon, ~ 0.3 cm hollow vase).

Regime comparison. The two regimes describe physically different materials, not two implementations of the same one. Surface-transmittance is correct for thin glazings and TIR-dominated geometry where the radiance is multiplied by τ at each crossing. Beer–Lambert is correct for bulk-absorbing glass where the attenuation grows continuously with the optical path. On a solid OG550 dragon (~ 1 cm), the Beer–Lambert regime is the physically faithful one and yields RGB-vs-codec ratios of $1.6\times$ to $6.0\times$ across the three Schott glasses of Sec. 6.2 of the main paper. The surface-transmittance regime applied to the same geometry produces larger codec gains ($8.22\times$ for OG550), but is a thin-filter approximation rather than a model of bulk absorption; the main paper reports both.

Implementation. In the surface-transmittance regime the latent codes are supplied to Mitsuba’s unmodified dielectric BSDF as `<rgb name="specularTransmittance">`, or to the homogeneous-medium plugin as a per-channel σ_a triplet in the Beer–Lambert regime, in each of the $k/3$ rendering passes.

6.1 Conductors: η , κ encoding and Fresnel accuracy

A conductor's reflectance is determined by its complex index of refraction $\eta(\lambda) - i\kappa(\lambda)$, which enters the Fresnel equations to produce the wavelength-dependent reflection coefficients at each surface interaction. Both η and κ are smooth spectral functions of the same form as a reflectance (measured by ellipsometry over 400–700 nm), so the codec encodes them with no class-specific machinery:

$$z_\eta = E(\eta), \quad z_\kappa = E(\kappa). \quad (7)$$

The non-linearity of the Fresnel equations in η, κ is the operation we cannot exactly commute with the encoder (the impossibility theorem, Sec. 1), but it acts *independently per wavelength*: applying Fresnel to the latent triplets in each pass produces the per-channel Fresnel of the encoded η, κ , with a small commutation error that the codec's training objective is designed to suppress. Empirical error stays below $2.2 \Delta E_{2000}$ (dragon-masked) across copper, gold, silver, aluminium, and chromium under D65.

Implementation. The latent triplets are passed to Mitsuba's unmodified rough-conductor BSDF as `<rgb name="eta">` and `<rgb name="k">` in each of the $k/3$ passes; no plugin modification is required.

Mirror approximation alternative. In the limit of a perfect mirror the Fresnel reflectance is angle-independent and equal to the spectral $|\eta - i\kappa - 1|^2 / |\eta - i\kappa + 1|^2$ at normal incidence. Replacing the conductor BSDF with a flat mirror whose spectral reflectance is encoded once gives a codec-friendly single-multiplication shortcut. We use the full roughconductor in the experiments to keep the comparison with spectral baselines fair (the latter have access to angle-dependent Fresnel); the mirror shortcut is recommended for downstream RGB rasterisers that have no conductor primitive of their own.

Per-material breakdown. Copper under D65 and three narrowbands appears in Tab. 7; per-metal dragon-masked means under D65 for all five measured conductors are reported in Tab. 4 of the main paper, and stay below $2.2 \Delta E_{2000}$. The narrowband-475 cell is the codec's hardest case across all materials and is analysed in Sec. 5.

6.2 Homogeneous subsurface scattering

A homogeneous participating medium attenuates and scatters light according to two wavelength-dependent coefficients: an absorption $\sigma_a(\lambda)$ and a scattering $\sigma_s(\lambda)$, with extinction $\sigma_t = \sigma_a + \sigma_s$. The radiance along a ray segment of length d accumulates per-channel transmittance $\exp(-\sigma_t d)$, and at each scattering event the single-scatter albedo $a(\lambda) = \sigma_s / \sigma_t$ multiplies the outgoing radiance. All three quantities are smooth spectral functions of the same form as the reflectances the encoder was trained on; we encode each independently:

$$z_{\sigma_a} = E(\sigma_a), \quad z_{\sigma_s} = E(\sigma_s), \quad z_a = E(a).$$

The volume-rendering equation processes each wavelength independently: extinction is a per-channel multiplication and the albedo is a per-channel multiplier. Replacing the three RGB channels that an unmodified renderer expects with the $k/3$ latent triplets therefore requires no plugin change: the same volumetric integration code path runs $k/3$ times, and the decoder recombines the result. Multi-bounce SSS evaluates the latent albedo's repeated self-product z_a^N

for N scattering events; this is the most demanding test of $\mathcal{L}_{\text{code}}$, and the self-product chain of Sec. 2.3 has a median error of $0.003 \Delta E_{2000}$ at $N=10$.

Scalar-density sampling of free paths. Volume integrators frequently sample the next interaction distance from a single scalar density (the "maximum" or "mean" free-path strategy), then weight the per-channel contribution by the relative transmittance at that distance. When $\sigma_t^{(p)}$ varies across the latent channels by less than roughly an order of magnitude (as it does for the spectral skin and milk profiles in our experiments), the scalar approximation introduces only a small bias relative to per-channel sampling, and we use the mean of the latent channels as the scalar. At higher σ_t ratios this bias becomes visible and true per-channel sampling is preferable; both options are available in standard renderers without plugin modification.

6.3 Heterogeneous subsurface scattering

When a participating medium has spatially varying optical properties, the spectral albedo $a(\mathbf{x}, \lambda)$ depends on position $\mathbf{x}$ as well as wavelength. The codec compresses the wavelength axis at every voxel independently (the same $E: \mathbb{R}^{47} \rightarrow \mathbb{R}^k$ runs in parallel across the $N_x \times N_y \times N_z$ spatial grid):

$$z_a(\mathbf{x}) = E(a(\mathbf{x}, \lambda)) \in \mathbb{R}^k, \quad \forall \mathbf{x}.$$

The result is a 4D tensor of latent voxels with k channels rather than 47. Density is wavelength-independent in the BSSRDF model used here, so it remains a single scalar volume shared across all $k/3$ rendering passes.

The volume-rendering equation processes each wavelength of the heterogeneous albedo identically to the homogeneous case: per-channel extinction sampling, per-channel transmittance accumulation along ray segments, and per-channel albedo multiplication at scattering events. Replacing 3 RGB albedo channels with k latent channels therefore requires no integrator modification, and the disk footprint drops by roughly $47/k$ (~ 50 MiB vs. ~ 390 MiB at $k=6$ on the $210 \times 106 \times 95$ vase grid).

Implementation. Each pass's 3-channel latent volume is written as a separate Mitsuba 0.6 VOL file and read at render time through the unmodified gridvolume plugin. The renderer's heterogeneous-medium integrator (Woodcock tracking with per-channel extinction sampling) processes each pass identically; pass outputs are recombined by the decoder.

Hand scene. The hand is a per-voxel spectral-albedo + density volume rendered through Mitsuba's heterogeneous-medium plugin. Density is obtained directly from the hand mesh by voxelisation; only the albedo carries spectral content. Authoring a spectral reflectance at every voxel by hand is impractical, so we build the albedo around the observation that a hand's appearance reduces to a small set of anatomical features (a baseline skin tone modulated by freckles, knuckles, fingertips, and veins) blended over the geometry. Each feature is authored as a single target sRGB colour together with a spatial influence field on the volume, and the per-voxel RGB albedo is their convex combination. Linearity then carries this

construction to the spectral side: if each feature target is independently lifted to a spectral reflectance, the per-voxel spectral albedo is the same convex combination of those five spectra. This reduces the spectral-upsampling problem from “millions of voxels” to “five colours”.

We lift those five colours using the skin reflectance LUT of Gitlina et al. [2020], which spans drained to dark skin tones; for each we pick the LUT entry whose D65-integrated sRGB is closest to the target in CIELAB, take that entry’s measured reflectance as the spectral shape, and rescale it per channel so the D65 integration returns the target sRGB exactly: physically motivated skin spectra, exact RGB anchor. The resulting per-voxel spectral albedo is then handed to the codec, one encode per voxel.

Vase scene. In the hollow-glass vase scene (Fig. 6), the Nero Marquina marble pedestal beneath the vases carries per-voxel spectral measurements. The marble’s per-voxel σ_a, σ_s profile was authored from a published measured BSSRDF tabulation; the codec encodes the spectral voxel data once per voxel; the encoder runs in ~ 3 s on the full grid on a single CPU thread. Glass-pixel ΔE_{2000} relative to the spectral ground truth is 1.79 (codec) vs. 3.94 (RGB baseline), a $2.2\times$ improvement.

Solid vs. hollow geometry. The same encoding strategy applied to the hollow vases (3 mm wall thickness, ~ 0.3 cm optical path through coloured material) yields the $2.2\times$ gain above; on a solid coloured-glass dragon (1 cm optical path) the gain rises to $6.02\times$ on the strongest example (Schott OG550), with $1.6\times$ to $6.0\times$ across the three Schott glasses of the main paper. The path-length dependence is expected: the codec’s accuracy advantage compounds across multiple spectral products (one per glass-medium crossing or volumetric scattering event), and longer optical paths see more crossings.

6.4 Extended material-class evaluation

The seven exemplars in Fig. 6 of the main paper were chosen to span the four material classes; the remaining five materials evaluated in Sec. 6.2 of the main paper are shown in Supp. Fig. 7. The visualisation pipeline is identical to the main-paper figure: brightness matching on the mask before computing ΔE_{2000} , a yellow-to-red colour map with uniform $v_{\max}=20$ across all rows, masked to the dragon silhouette.

6.5 Colour-error characterisation of the per-channel regimes

Each per-channel regime of Sec. 4 of the main paper replaces an exact spectral operation with a latent one. Fig. 8 isolates each regime and tracks its colour error as the operation compounds, evaluated exactly as the pipeline operates: every latent chain is decoded jointly with the encoded D65 illuminant and scored with the per-spectrum metric of Sec. 6.3 of the main paper (the illuminant’s own white point). The green bands and markers overlay the rendered scene means of Sec. 6.2 of the main paper, so each stress curve can be read against the corresponding average over real path distributions. The rendered means sit well below the curve tips: real paths mix crossing counts, depths, and spectra, whereas each curve compounds

a single operation on a single spectrum. The curves are per-regime envelopes, not predictions of scene error.

(a) *Products and powers.* A path that crosses the same filter N times multiplies its spectrum N times; the panel tracks the latent self-product error for the five encoded Schott transmittances (the three rendered as dragons in the main paper plus GG495 and VG9; all five appear as hollow vases in Fig. 6). The dashed curve repeats the experiment for the mean test reflectance: repeated reflections in a multi-bounce path multiply reflectances in exactly the same way, so this curve is a synthetic worst case of multi-bounce transport in which every bounce hits the same average surface. The green band marks the measured rendered multi-bounce matpreview means (0.91–1.32, main-paper Sec. 6.2): real paths mix different surfaces whose encoding errors partially cancel, which is why the rendered means sit at the foot of the stress curves. Error grows slowly and irregularly rather than compounding, staying below $5 \Delta E_{2000}$ through twelve crossings.

(b) *Exponential sums.* The Beer–Lambert chain raises the encoded transmittance to the power d/d_{ref} per channel. The chain terminates on the mean test reflectance (a transmitted path always ends on a surface), so the $d=0$ intercept equals the codec’s product-error floor of 0.96; the dotted line marks the reference depth used by the encoding. The markers are the rendered glass-scene means, which average over each scene’s path-length distribution and therefore sit below the fixed-depth curves.

(c) *Per-channel Fresnel.* Colour error of the Fresnel response evaluated per channel on the encoded (η, κ) against the true spectral Fresnel, per incidence angle: nearly constant below 60° , tracking the reconstruction error of the encoded pair, and adding about two units at grazing incidence. The band spans the rendered dragon-masked means of the five measured conductors.

(d) *Ratios.* The naive per-channel single-scatter albedo, $z_s \oslash (z_a + z_s)$ ($\oslash$: per-channel division), raised to the scattering order. The smooth subsurface materials stay below $6 \Delta E_{2000}$ through twenty orders; wine’s saturated absorption makes its albedo strongly chromatic and the naive ratio shows the largest deviation of the set at low orders (beyond $n \approx 4$ both the true and the approximated albedo powers are effectively black and the difference vanishes), which is why such media use the factorised encoding of Sec. 4 of the main paper. The markers show the rendered homogeneous-SSS scene means and wine’s rendered result under the factorised encoding.

7 Upsampling networks for legacy RGB assets

Two separate networks map $\text{RGB} \rightarrow 6\text{-D latent}$: a reflectance upsampler and an illumination upsampler. Both are small bias-free MLPs in linear-XYZ space, magnitude-equivariant ($f(\alpha \text{xyz}) = \alpha f(\text{xyz})$): the network operates on the unit direction; magnitude is reattached at the output.

Architectures. The reflectance network has a dual head: a shared trunk $\text{Linear}(3, 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128, 64) \rightarrow \text{ReLU}$, a latent head $64 \rightarrow k$, and an auxiliary spectrum head $k \rightarrow 128 \rightarrow 128 \rightarrow 47$ used only for evaluation. The illumination network is single-head: $\text{Linear}(3, 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, 32) \rightarrow \text{ReLU} \rightarrow \text{Linear}(32, k)$.

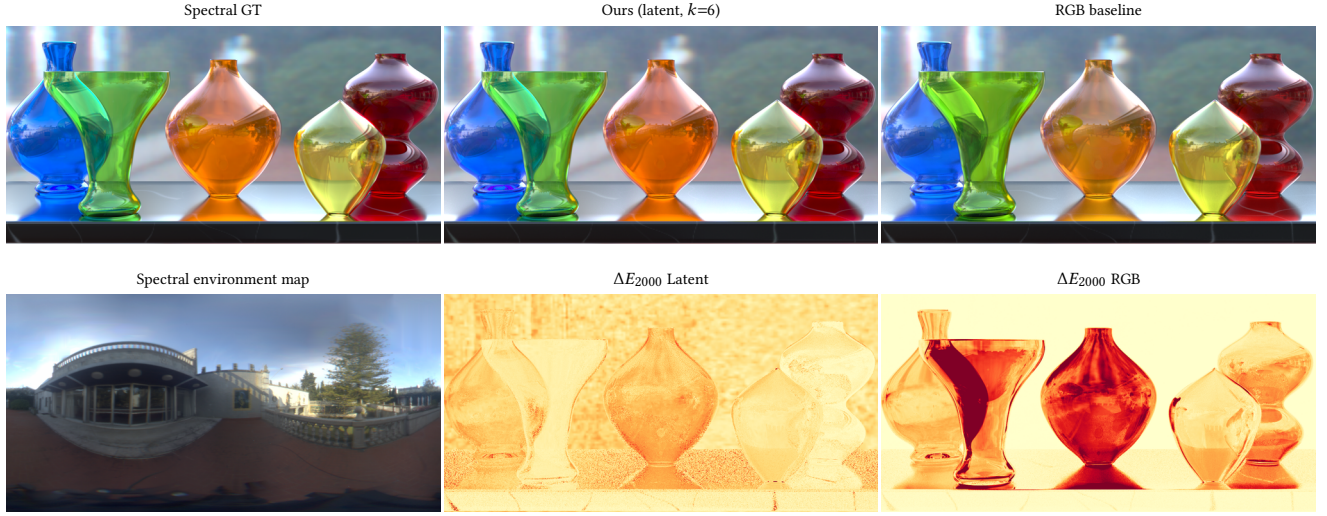


Fig. 6. **Marble-pedestal vase scene (heterogeneous SSS)**. Top row: spectral GT (CMF-integrated to sRGB), our latent reconstruction at $k=6$ (two RGB passes), and the standard RGB baseline through the same renderer. Bottom row: the spectral environment map used as illumination (the envmap’s own spectral slices, CMF-integrated for display), and per-pixel ΔE_{2000} for the latent and the RGB baseline against the spectral GT. Hyperspectral environment map from Morimoto et al. [2019]. Glass-pixel $\Delta E_{2000}=1.79$ for the codec vs. 3.94 for the RGB baseline, a $2.2\times$ improvement on the hollow geometry (3 mm wall, ~ 0.3 cm optical path through coloured material).

Training. Loss for reflectance:

$$\mathcal{L}_R = \lambda_{\text{lat}} \Phi(\hat{z}, E(R)) + \lambda_{\text{spec}} \text{MSE}(\hat{s}, R) + \lambda_{\text{col}} \Delta E_{76}(\hat{s}, R) + \lambda_{\text{smooth}} \|\hat{s}_{i+1} - \hat{s}_i\|^2,$$

with $(\lambda_{\text{lat}}, \lambda_{\text{spec}}, \lambda_{\text{col}}, \lambda_{\text{smooth}}) = (1.0, 1.0, 0.5, 0.1)$. Illumination: $\mathcal{L}_L = \Phi(\hat{z}, E(L))$ only. Adam, learning rate 10^{-3} , batch size 256, 300 epochs, early stop at patience 30. Train/val/test splits mirror the codec’s (Sec. 3.3).

Held-out evaluation. Tab. 10 reports held-out test error for the two networks against the direct codec on the same spectra. FL11 is the

Table 10. Held-out test ΔE_{2000} . “upsampler” encodes from RGB; “direct codec” encodes from the original spectral input.

	upsampler	direct codec
reflectance (218 spectra)	1.18	0.85
illumination (4 representatives)	1.31	1.15

worst-case illuminant: the network smooths the spike structure of its multi-peaked spectral power distribution (SPD), raising the 95th-percentile ΔE_{2000} to 5.38 versus 3.17 for the direct codec. Predicting the k -dimensional latent rather than the full n -dimensional spectrum is what makes the codec-as-target formulation work: the smaller, smoother target matches the upsampler’s capacity.

Spectral-domain fidelity of uplifted illuminants. The illumination-upsampling comparison in the main paper (Sec. 6.3) evaluates every method through the codec, as used in the pipeline. For reference, Tab. 11 reports the fidelity of the uplifted illuminant spectra themselves, before encoding: the uplifted spectrum multiplies the ground-truth reflectances directly. The scaled uplift reproduces D65 almost

exactly, since its RGB round-trip is a hard constraint; every method then loses accuracy once its spectrum passes through the codec, which the main-paper table includes.

Table 11. Spectral-domain ΔE_{2000} (mean / p95) of upsampled illuminants over the 218 held-out reflectances, without codec involvement. Same fits as the main-paper rows.

Method	D65	LED-B1	LED-B3	FL11
Scaled uplift (J&H×D65)	0.06 / 0.11	1.18 / 4.11	1.43 / 3.58	1.48 / 4.68
Guarnera et al., LED	2.13 / 4.23	4.49 / 10.34	1.49 / 3.35	1.40 / 4.40
Guarnera et al., Gaussian	2.31 / 6.63	3.09 / 6.63	1.59 / 3.46	2.64 / 5.47

Directional error. A route can be accurate on average yet push colour consistently in one direction. We therefore report the *mean signed* CIELAB offset, the average of the signed $(\Delta L^*, \Delta a^*, \Delta b^*)$ vectors over the 218 held-out reflectances, whose magnitude is zero when errors cancel and large only when they share a direction. Tab. 12 gives it for the reflectance routes. Substituting our upsampler for encoded ground-truth spectra changes the offset by at most 0.12; the uplift-then-encode route’s offset is larger on all four illuminants.

To test that increment rather than assert it, we treat the no-upsampling reference as the paired control and bootstrap the difference. Because many test reflectances share a spectral shape, resampling 33 shape-correlated groups (single linkage at cosine 0.995; the largest holds 149 of the 218 spectra), a deliberately conservative unit. Against an equivalence margin of 0.5 CIELAB units the increment is decisively small under D65 and LED-B3 and inconclusive under LED-B1 and FL11, where the interval is wide rather than the point estimate large.

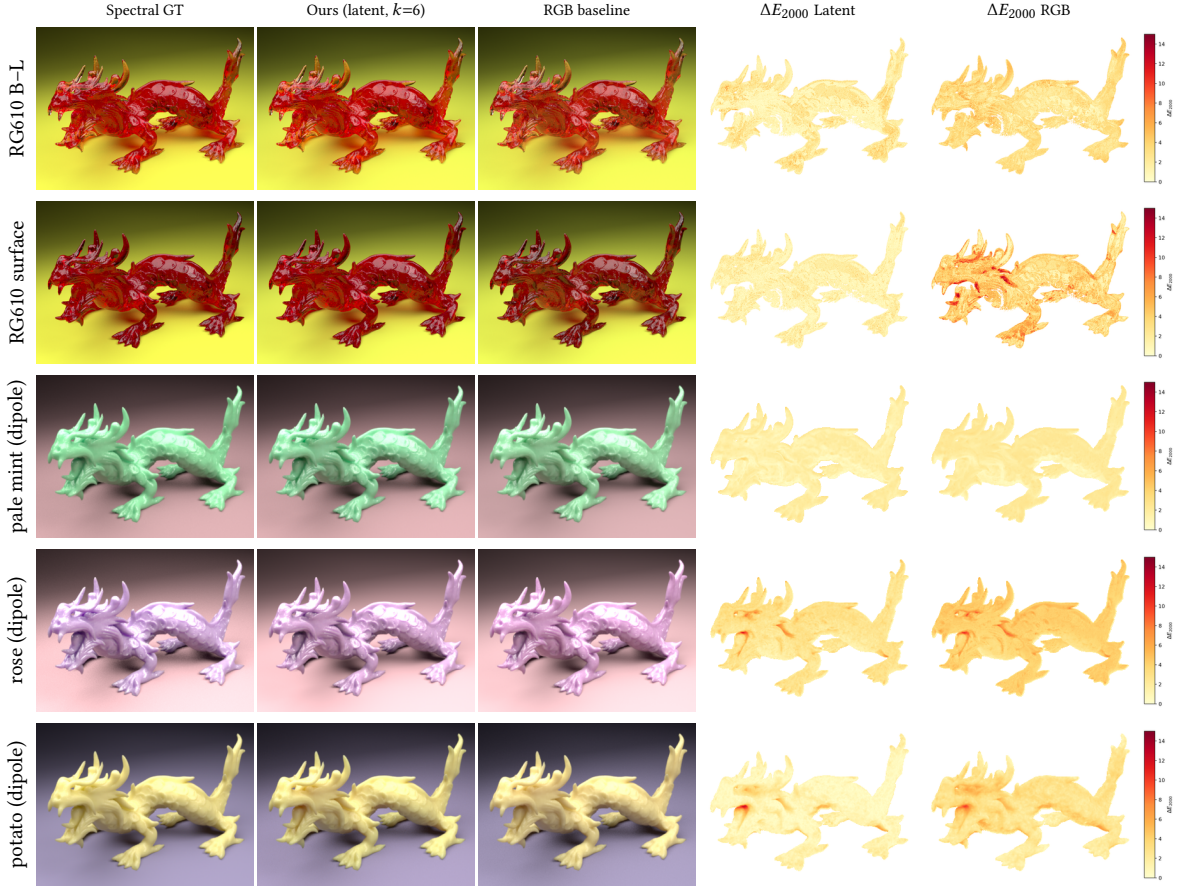


Fig. 7. **Material-class evaluation: additional materials.** Same column layout, codec configuration, and visualisation pipeline as Fig. 6 of the main paper: latent $k=6$, ΔE_{2000} versus spectral GT, per-row colour bar to the right of the ΔE_{2000} RGB panel. The seven main-paper exemplars cover the most demanding case in each material class; this figure shows the remaining glasses and dipoles that round out the per-class numerical evaluation in Sec. 6.2 of the main paper. Row labels: “B–L” is the Beer–Lambert volumetric-absorption regime and “surface” the surface-transmittance regime of Sec. 6.

The illumination upsampler is reported on the 48-lamp population of Tab. 6 of the main paper, where each lamp is an independent condition. Its increment is 0.342 with a 95% interval of [0.213, 0.631], which crosses the margin, so on this evidence its directional behaviour is unresolved: it is neither shown to be free of directional error nor shown to carry one.

8 Comparison against a modern spectral renderer

We benchmark our codec against Mitsuba 3, a modern spectral renderer whose path tracer implements hero wavelength sampling [Wilkie et al. 2014], in which each ray carries a stochastically chosen “hero” wavelength together with companion wavelengths stratified across the visible band, and the integrated radiance is estimated by Monte Carlo across rays. Mitsuba 3’s spectral variants sample four wavelengths per ray with the shifted-stratified construction of Wilkie et al. [2014], mapped through the sensor response curve when one is set.

Our approach is conceptually orthogonal (hero wavelength sampling is a sampling strategy inside a spectral renderer, while our

Table 12. **Mean signed CIELAB offset** (magnitude of the average signed Lab vector) over the 218 held-out reflectances, and the paired increment of the reflectance upsampler over the no-upsampling reference with its 95% cluster-bootstrap interval. Lower is a smaller directional bias; the offset is not an error magnitude and is not comparable with ΔE_{2000} in Tab. 5 of the main paper.

	D65	LED-B1	LED-B3	FL11
<i>Mean signed offset</i>				
Jakob & Hanika (spectral)	0.45	0.28	0.24	0.64
J&H $\rightarrow$ codec	1.99	0.98	1.28	1.75
Ours, $f^{(R)}$	0.90	0.63	0.77	1.06
Reference (codec on GT)	0.94	0.61	0.79	0.97
<i>Paired increment of $f^{(R)}$ over the reference</i>				
increment	0.079	0.022	0.028	0.122
95% CI	[0.03, 0.27]	[0.02, 0.85]	[0.02, 0.42]	[0.07, 1.23]
verdict (margin 0.5)	equiv.	inconcl.	equiv.	inconcl.

codec replaces the spectral renderer with $k/3$ passes of an unmodified RGB renderer), but the natural empirical question is how the

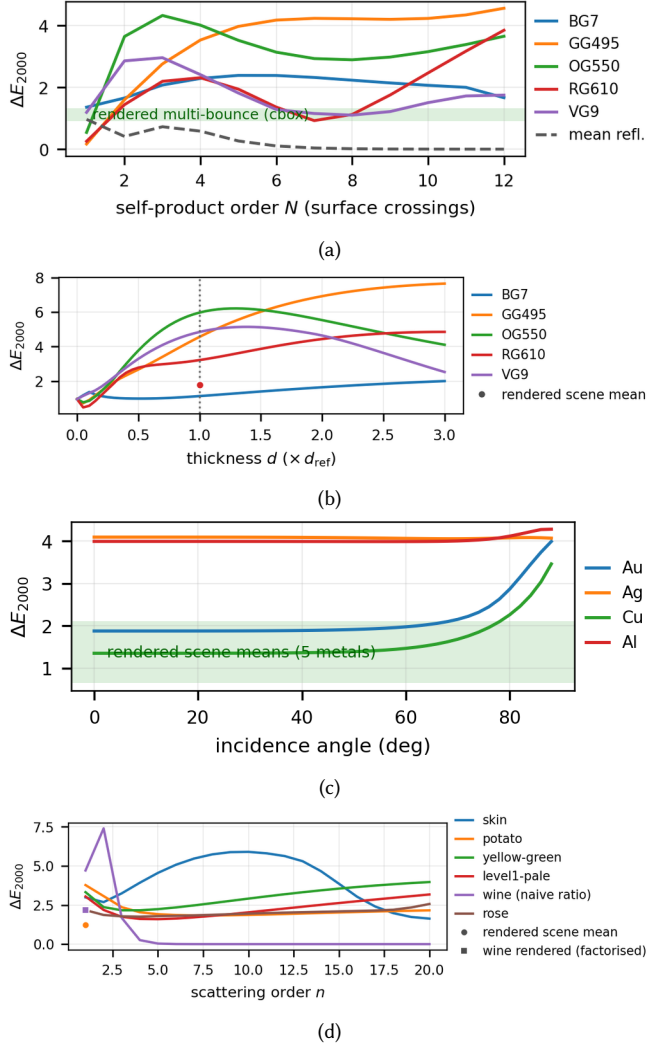


Fig. 8. **Colour error of the four per-channel regimes**, evaluated as the pipeline operates: each latent chain is decoded jointly with the encoded D65 illuminant and scored with the per-spectrum metric of Sec. 6.3 of the main paper. Green bands and markers show the corresponding rendered scene means of this section. (a) Repeated Hadamard self-product of encoded Schott transmittances and the mean test reflectance: error grows slowly with the number of crossings instead of compounding. (b) Beer-Lambert transmittance deviation versus optical depth; the chain terminates on the mean test reflectance, so the $d=0$ intercept is the codec's product-error floor (0.96); the dotted line marks the reference depth d_{ref} used by the encoding. (c) Per-channel conductor Fresnel deviation versus incidence angle: nearly constant below 60° , tracking the reconstruction error of the encoded (η, κ) . (d) Per-channel single-scatter albedo versus scattering order: bounded for the smooth subsurface materials; wine's saturated bands make its naive ratio the largest deviation of the set at low orders (beyond $n \approx 4$ both the true and the approximated albedo powers are effectively black and the colour difference vanishes). Such media use the factorised encoding of Sec. 4 of the main paper.

latent two-pass pipeline compares in wall-clock against a renderer that uses hero wavelength sampling, at matched resolution and spp. We test on the standard cbox scene under broadband illumination, sweeping spp and image resolution. Both runs use Mitsuba 3 on the same machine, single-threaded, without GPU acceleration: the spectral mode exercises hero wavelength sampling, while the latent mode drives the same renderer in two RGB passes.

Timing parity. All reported latent-pipeline times are *end-to-end*: they sum the codec encode of the matched scene inputs, both RGB render passes, and the codec decode that reconstructs the spectral image from the two latent passes. Across all 15 tested configurations, the latent two-pass pipeline lands within a median of about +5% wall-clock of Mitsuba 3's spectral output, and within $\pm 10\%$ on 14/15 configurations (Tab. 13; the wall-clock curves are essentially indistinguishable on log-log axes, Fig. 9). Encode and decode contribute a negligible fraction of that end-to-end cost: encoding is a single matrix multiply on the framebuffer (< 10 ms); decoding runs at 0.005–0.5 s and stays below 0.5% of the total render time at 1024 spp and above. The lower blocks of Tab. 13 show the scene dependence of this parity. On the Grey & White Room (65 meshes, textured materials, environment lighting through windows) hero wavelength sampling costs only about $1.2\times$ an RGB render, so the fixed two-pass cost of the latent pipeline exceeds the spectral wall-clock by 75–81%; adding three FL11 area lamps to the same scene raises the spectral per-sample cost and narrows the excess to 63–73%. On shading-bound scenes the two pipelines approach parity: +13–29% on Veach MIS and +13–22% on the Glass of Water still life, bracketing the Cornell-box results above. Asset encoding (texture, environment-map, and constant upsampling) is a one-off preprocess and is excluded from the per-frame times; decode is included.

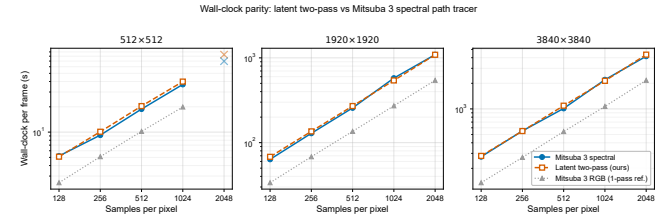


Fig. 9. Wall-clock per frame as a function of samples per pixel, one panel per image resolution (log-log). Mitsuba 3 spectral (solid blue) and our latent two-pass (dashed orange) lie almost on top of each other across the full 128–2048 spp range; the single-pass Mitsuba 3 RGB reference (dotted grey) sits at roughly half the cost as expected. The faded $\times$ markers in the 512² panel mark the 2048 spp configuration, the one case outside $\pm 10\%$.

Quality at matched spp. At equal spp, the latent two-pass renders are consistently more accurate than the RGB-only baseline and approach the Mitsuba 3 spectral reference from a different direction. The spectral reference's residual error decreases with spp (it is a Monte Carlo estimator), whereas the latent error is roughly *constant* in spp once framebuffer noise has decayed (the codec's reconstruction is deterministic). At low spp the latent approach has the lower ΔE_{2000} because it is noise-free; at high spp the spectral reference converges to its own ground truth and the gap closes. Across all

Table 13. End-to-end wall-clock per frame, latent two-pass vs. Mitsuba 3 spectral, across resolution and spp. Latent times are total end-to-end; the bracketed values split out codec encode and decode (decode only in the scene blocks). Overhead is $(t_{\text{lat}} - t_{\text{spec}}) / t_{\text{spec}}$. The $512^2 \times 2048$ spp row is the only configuration outside $\pm 10\%$.

Resolution	spp	Mitsuba 3 spectral (s)	Latent total (s) [enc, dec]	Overhead
512^2	128	5.20	5.10 [enc: 0.005, dec: 0.007]	-1.9%
512^2	256	9.20	10.10 [enc: 0.001, dec: 0.005]	+9.8%
512^2	512	18.80	20.40 [enc: 0.001, dec: 0.005]	+8.5%
512^2	1024	36.90	39.90 [enc: 0.001, dec: 0.005]	+8.1%
512^2	2048	69.80	83.50 [enc: 0.008, dec: 0.069]	+19.6%
1920^2	128	63.57	68.05 [enc: 0.007, dec: 0.065]	+7.0%
1920^2	256	128.92	135.85 [enc: 0.001, dec: 0.064]	+5.4%
1920^2	512	256.90	270.20 [enc: 0.001, dec: 0.066]	+5.2%
1920^2	1024	576.97	541.79 [enc: 0.001, dec: 0.069]	-6.1%
1920^2	2048	1091.47	1087.81 [enc: 0.001, dec: 0.070]	-0.3%
3840^2	128	274.10	280.20 [enc: 0.006, dec: 0.484]	+2.2%
3840^2	256	551.20	549.30 [enc: 0.002, dec: 0.321]	-0.3%
3840^2	512	1012.90	1090.90 [enc: 0.001, dec: 0.320]	+7.7%
3840^2	1024	2203.40	2137.50 [enc: 0.008, dec: 0.332]	-3.0%
3840^2	2048	4157.50	4344.80 [enc: 0.001, dec: 0.262]	+4.5%
<i>Grey & White Room interior (scene: Wig42, CC BY 3.0)</i>				
1920×1080	128	81.30	147.07 [dec: 0.028]	+80.9%
1920×1080	512	336.48	593.89 [dec: 0.034]	+76.5%
1920×1080	1024	678.73	1189.32 [dec: 0.031]	+75.2%
<i>Grey & White Room + three FL11 area lamps (controlled spectral load)</i>				
1920×1080	128	67.92	112.51 [dec: 0.028]	+65.7%
1920×1080	512	274.69	446.52 [dec: 0.035]	+62.6%
1920×1080	1024	521.43	903.99 [dec: 0.032]	+73.4%
<i>Veatch MIS (scene: Benedikt Bitterli, CC0)</i>				
1920×1080	128	29.11	33.03 [dec: 0.028]	+13.5%
1920×1080	512	102.39	132.08 [dec: 0.034]	+29.0%
1920×1080	1024	206.83	264.77 [dec: 0.033]	+28.0%
<i>Glass of Water (scene: aXel, CC0)</i>				
1920×1080	128	30.84	37.55 [dec: 0.027]	+21.8%
1920×1080	512	122.32	144.94 [dec: 0.029]	+18.5%
1920×1080	1024	256.46	289.99 [dec: 0.034]	+13.1%

configurations tested, the latent result has a lower ΔE_{2000} than the RGB-only baseline (Fig. 10 and Tab. 14).

Table 14. Mean ΔE_{2000} vs. Mitsuba 3 spectral GT, cbox under broadband; each range spans the three resolutions of Tab. 13.

spp	latent (ours)	RGB baseline	ratio
128	2.55–2.63	3.93–4.01	1.5×
256	2.09–2.19	3.50–3.55	1.7×
512	1.80–1.84	3.22–3.26	1.8×
1024	1.64–1.67	3.03–3.06	1.8×
2048	1.56–1.58	2.91–2.94	1.9×

A spectral renderer with hero wavelength sampling is preferred when the deployment target supports one; our codec is preferred when the deployment target is a closed-source RGB renderer that has no spectral primitive. On renderers that support both, the two are cost-neutral at low-medium spp.

9 Cross-renderer integration: additional material

Sec. 6.4 of the main paper demonstrates the codec deploying into four production rendering frameworks (Blender Cycles, Maya Arnold,

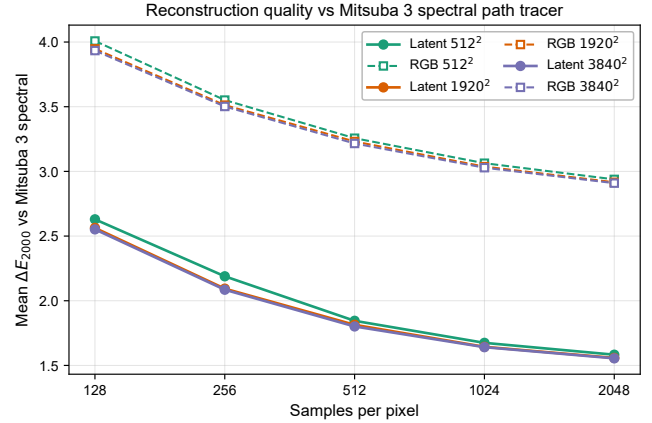


Fig. 10. Reconstruction quality: mean ΔE_{2000} vs. the Mitsuba 3 spectral reference for the latent two-pass (solid) and the Mitsuba 3 RGB baseline (dashed), as a function of samples per pixel and across image resolutions. The latent curves converge to $\sim 1.6 \Delta E_{2000}$ by 2048 spp (the codec’s residual reconstruction error), while the RGB baseline floor is roughly twice that ($\sim 2.9 \Delta E_{2000}$). Resolution has essentially no effect on either curve.

KeyShot, and Unreal Engine 5) with no per-pipeline modification, measured against a spectral reference constructed inside each renderer. This section gives the ground-truth construction protocol, the per-renderer configurations, and the complete per-case scores for all thirteen scene and lighting cases; the main paper shows eight of them.

Spectral ground truth by RGB-packed passes. Commercial renderers expose no spectral output, so the reference is built inside each renderer rather than imported from ours. Every material is uplifted to a reflectance spectrum with Jakob and Hanika [2019] and every light source uses its known SPD; no component of our codec enters the reference. The 47 channels covering 368 to 830 nm are rendered in 16 passes, each pass carrying three consecutive channels in the renderer’s R, G, and B slots. Passes are written as linear floating-point images and reassembled into a 47-channel cube, with a per-renderer response and exposure calibration applied so that all passes share one linear scale. Because the Jakob and Hanika model is smooth by construction, the reflectance side of this reference carries no sharp spectral features; the illumination side does, since the lights are taken from the measured library. The KeyShot environment-map case differs in kind: an RGB environment map carries no measured SPD, so its reference sky is the uplift of the map itself, and all three routes are evaluated against that reference. That map is an 8-bit image whose bright regions are clipped, which makes the reference sky spectrally flat wherever clipping occurs, so this case is the least spectrally demanding of the thirteen.

Per-renderer configuration.

- **Blender Cycles.** Path tracing integrator, 128 spp with built-in denoising. Two test conditions: broadband three-light setup (front key + side fill + top), and narrowband with three high-intensity coloured spotlights added above the

Table 15. **All thirteen cross-renderer cases:** mean / 95th-percentile ΔE_{2000} against each renderer's spectral reference for the codec on spectral assets, the codec with our RGB upsampler, and the RGB baseline. Cases marked * appear in the main paper.

renderer	case	codec, spectral assets	codec + our RGB upsampler	RGB baseline
Blender Cycles	narrowband*	1.37 / 2.82	1.40 / 2.94	4.22 / 8.44
	broadband*	1.53 / 2.33	1.47 / 2.27	3.80 / 6.00
Maya Arnold	light 0002*	1.87 / 2.68	1.57 / 2.83	0.69 / 1.84
	light 0099*	2.26 / 3.07	2.45 / 3.44	4.25 / 10.49
	light 0333	0.39 / 0.89	0.66 / 1.76	2.90 / 8.11
	light 0414	1.25 / 2.08	1.24 / 3.03	3.80 / 11.38
KeyShot	environment map*	2.58 / 5.50	3.80 / 9.73	8.56 / 20.91
	uniform environment 0099	2.74 / 3.99	3.82 / 7.65	4.84 / 13.25
	uniform environment 0333*	0.34 / 0.83	1.69 / 6.55	5.26 / 14.50
Unreal Engine 5	lights 492*	1.65 / 2.96	2.36 / 6.39	4.20 / 9.81
	cool lights*	1.59 / 3.50	2.08 / 6.87	4.22 / 12.65
	alternative lights 454	1.68 / 3.11	2.59 / 7.11	4.39 / 10.06
	strong soft shadow 441	1.94 / 5.39	3.45 / 9.49	5.58 / 13.74

scene, peaking at 659, 559, and 458 nm with FWHM of 52, 48, and 43 nm.

- **Maya Arnold.** Path tracing. Camera AA = 3 (9 primary samples per pixel), Diffuse = 2, Specular = 1, Transmission = 8. Four cases, one known-SPD light each.
- **KeyShot.** Built-in path tracer at 128 spp. Three cases: the scene's own environment map, and two uniform-colour environment lights carrying the same known SPDs as the Maya cases.
- **Unreal Engine 5.** Real-time rasteriser with Lumen global illumination (not path tracing). Front area light + side fill light + top spot light, four lighting sets.
- **Reference (Mitsuba).** Main-paper Mitsuba renderings use path tracing at the sample counts stated in each figure caption (512 to 4,096 spp); ground-truth comparisons use Mitsuba 0.6's spectral build.

All thirteen cases. Mean and 95th-percentile ΔE_{2000} against the per-renderer spectral reference (Tab. 15). RGB-pipeline renders are brightness-matched to the reference before comparison; without that match they score 28 to 51, which measures an exposure and unit mismatch rather than a colour error.

Cases marked * appear in the main paper; Figs. 11–15 show the remaining five. The codec's own error stays between 0.34 and 2.74 across every renderer and lighting condition, and the upsampling premium stays bounded, at most 1.5 ΔE_{2000} above the codec except under Unreal's strongest metameric set. The RGB baseline is worse than both routes in twelve of the thirteen cases. The exception is Maya's light 0002, a nearly flat broadband source under which an RGB pipeline has almost nothing to get wrong; the SPD insets in Fig. 11 and in Fig. 11 of the main paper show how that light compares with the structured ones.

Temporal stability. We render 120-frame animated sequences in Blender Cycles under both broadband and narrowband lighting (camera and light animation), with no per-frame codec retraining. A single combined supplemental video shows both segments back-to-back: a broadband title card, the 120-frame broadband sequence, a narrowband title card, then the 120-frame narrowband sequence.

Every video frame inside the sequences is a three-panel composite at 1080 px per panel (*Standard RGB* / *Codec (Ours)* / *Signed error*), with the per-pixel signed error computed in linear sRGB. The video is encoded at 15 fps. No flicker, chromatic drift, or temporal inconsistency is visible across the 120-frame range under either condition.

References

- Michael F. Atiyah and Ian G. Macdonald. 1969. *Introduction to Commutative Algebra*. Addison-Wesley, Reading, MA.
- Yuliya Gitlina, Giuseppe Claudio Guarnera, Daljit Singh Dhillon, Jan Hansen, Alexandros Lattas, Dinesh Pai, and Abhijeet Ghosh. 2020. Practical Measurement and Reconstruction of Spectral Skin Reflectance. *Computer Graphics Forum* 39, 4 (2020), 75–89. doi:10.1111/cgf.14055
- Wenzel Jakob and Johannes Hanika. 2019. A Low-Dimensional Function Space for Efficient Spectral Upsampling. *Computer Graphics Forum* 38, 4 (2019), 147–155. Proceedings of Eurographics Symposium on Rendering. doi:10.1111/cgf.13626
- Diederik P. Kingma and Jimmy Ba. 2014. Adam: A Method for Stochastic Optimization. arXiv:1412.6980 [cs.LG]
- LSPDD. 2026. Lamp Spectral Power Distribution Database (LSPDD). <https://lspdd.org/>. Accessed: 2026-01-22.
- Takuma Morimoto, Sho Kishigami, João M. M. Linhares, Sérgio M. C. Nascimento, and Hannah E. Smithson. 2019. Hyperspectral Environmental Illumination Maps: Characterizing Directional Spectral Variation in Natural Environments. *Optics Express* 27, 22 (Oct 2019), 32277–32293. doi:10.1364/OE.27.032277
- Poorvi L. Vora and H. Joel Trussell. 1993. Measure of goodness of a set of color-scanning filters. *Journal of the Optical Society of America A* 10, 7 (1993), 1499–1508. doi:10.1364/JOSAA.10.001499
- Alexander Wilkie, Sehera Nawaz, Marc Droske, Andrea Weidlich, and Johannes Hanika. 2014. Hero Wavelength Spectral Sampling. *Computer Graphics Forum* 33, 4 (2014), 123–131. Proceedings of Eurographics Symposium on Rendering. doi:10.1111/cgf.12419

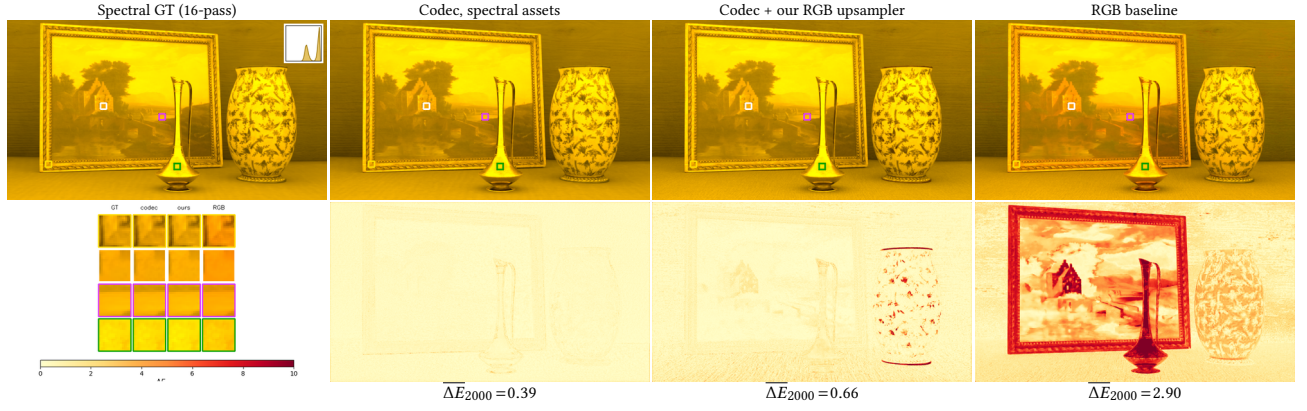


Fig. 11. **Maya Arnold, light 0333**. Layout as in the main paper: top row the three routes against the 16-pass spectral reference, bottom row the magnified patches with the shared colour bar and the per-pixel ΔE_{2000} maps (scale capped at 10). The inset in the reference cell shows the light SPDs.

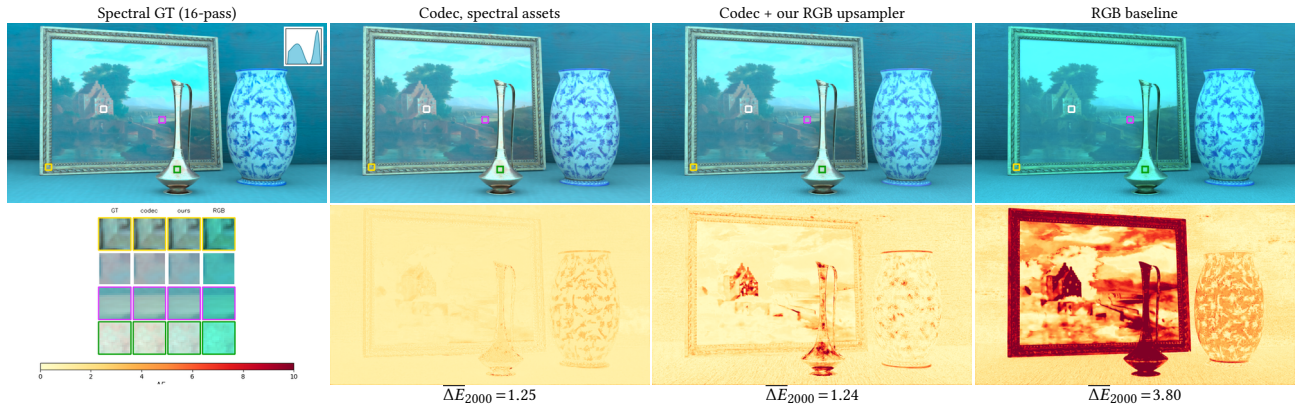


Fig. 12. **Maya Arnold, light 0414**. Layout as in the main paper: top row the three routes against the 16-pass spectral reference, bottom row the magnified patches with the shared colour bar and the per-pixel ΔE_{2000} maps (scale capped at 10). The inset in the reference cell shows the light SPDs.

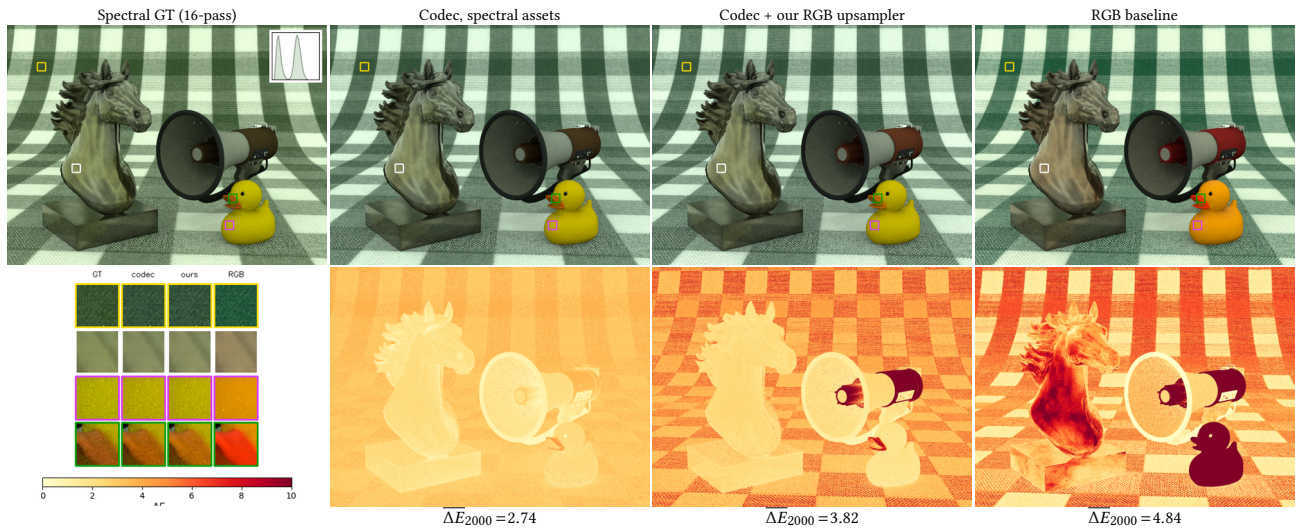


Fig. 13. **KeyShot, uniform environment light 0099**. Layout as in the main paper: top row the three routes against the 16-pass spectral reference, bottom row the magnified patches with the shared colour bar and the per-pixel ΔE_{2000} maps (scale capped at 10). The inset in the reference cell shows the light SPDs.

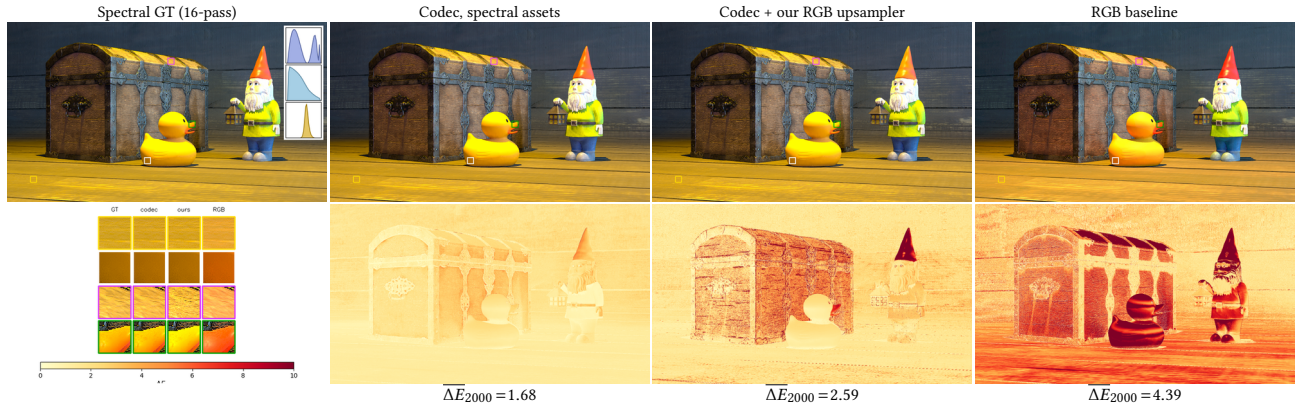


Fig. 14. **Unreal Engine 5, alternative lights 454.** Layout as in the main paper: top row the three routes against the 16-pass spectral reference, bottom row the magnified patches with the shared colour bar and the per-pixel ΔE_{2000} maps (scale capped at 10). The inset in the reference cell shows the light SPDs.

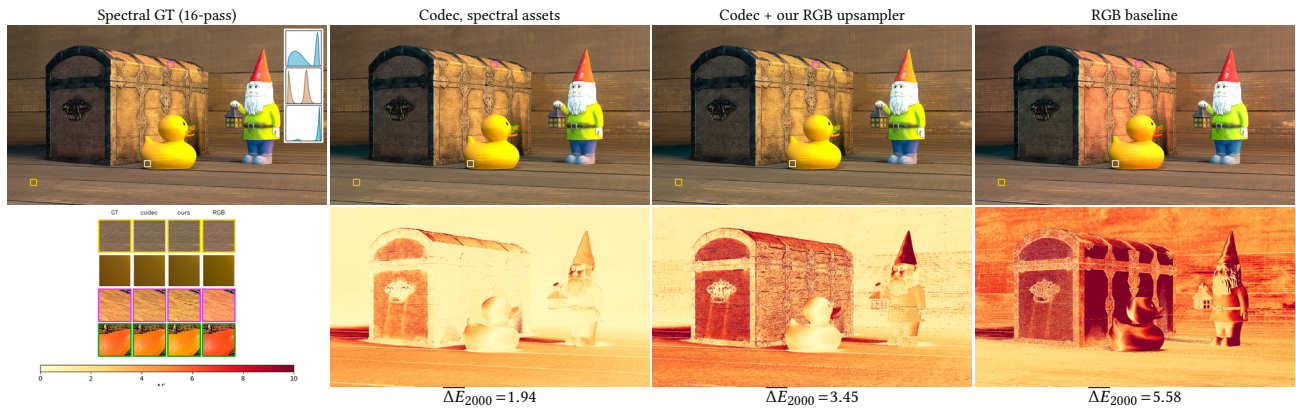


Fig. 15. **Unreal Engine 5, strong soft shadow 441.** Layout as in the main paper: top row the three routes against the 16-pass spectral reference, bottom row the magnified patches with the shared colour bar and the per-pixel ΔE_{2000} maps (scale capped at 10). The inset in the reference cell shows the light SPDs.